Pharmacy Capacity and the NHS 60-Day Clinical-Trial Set-up Target

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Abbreviations

C&C	Capability and Confirmation Agreement
CTIMP	Clinical Trial of an Investigational Medicinal Product
ePMA	Electronic Prescribing and Medicines Administration
GMO	Genetically Modified Organism (committee)
HRA	Health Research Authority
IDEAS	An LTHTR research review committee
LTHTR	Lancashire Teaching Hospitals NHS Foundation Trust
M/G/1	Markovian arrivals, general service, single server
NHS	National Health Service
NIHR	National Institute for Health and Care Research
PPCTI	LTHTR's aseptic pre-trial sign-off checklist
R&I	Research and Innovation (LTHTR department)
RQ, SQ	Research question, sub-question

Abstract

The United Kingdom has experienced a decline in commercial clinical-trial activity, while national policy requires regulatory application to first participant recruitment within 150 days, including a 60-day site set-up window. Pharmacy approval is a necessary condition for site opening. At Lancashire Teaching Hospitals, only 3 of 24 commercial trials met this target during 2024–2026, with a median site set-up time of 144 days, indicating a substantial challenge in allocating constrained pharmacy set-up capacity.

This study asks how many commercial medicine trials the trust's existing pharmacy capacity can set up on time, and what limits that number. It develops and calibrates a planning model that translates pharmacy clinical-trial set-up, pathway requirements, and the shared approval step into transparent monthly commitment decisions. The binary integer programme selects trials lexicographically, maximising on-time completions first and minimising weighted lateness among tied plans. Final pharmacy approval is represented as an M/G/1 shared queue whose expected delay increases with the number of trials committed. The model was calibrated using 20 completed case studies and a 104-trial operational tracker, replayed across 25 historical decision points, and tested against simulated portfolios.

Only four distinct trials were ever feasible for on-time completion. Raising the monthly commitment limit added no on-time completions at any of the 25 decision points; from three commitments onwards, the model selected the same trials as the arrival-order comparator. Selection could therefore reallocate scarce capacity but could not recover time lost before pharmacy received the work. The binding constraint was usable time rather than nominal pharmacy capacity, because a high-burden set-up could consume the entire remaining window alone.

Sensitivity analysis showed that preparation before the target clock started was the only tested intervention to raise the on-time ceiling. Reducing hand-over and pharmacy-queue delays improved trials near the threshold and reduced lateness, whereas additional capacity reduced lateness without increasing on-time completions. The study therefore provides a calibrated decision-support model showing that portfolio selection must complement earlier preparation and delay reduction, rather than substitute for them.

Declaration of Authorship

I, **Kachanon Lertdamrongwong**, hereby declare that this thesis entitled, **"Pharmacy Capacity and the NHS 60-Day Clinical-Trial Set-up Target"**, is all my own work, except as indicated in the text.

No portion of the work referred to in the dissertation has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

Signed:
Kachanon Lertdamrongwong

Date:
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Chapter 1

Introduction

1.1 Background and motivation

Between 2017 and 2022, O'Shaughnessy (2023) reports, the United Kingdom fell from fourth to tenth in the world for the number of commercial clinical trials initiated, and enrolments in commercially sponsored NIHR studies fell by 44 per cent, from roughly 50,000 patients a year to 28,000. The decline matters because patients lose access to treatments available only through trial participation.

The Department of Health and Social Care introduced a national 150-day metric, from regulatory application to first participant recruitment, to improve the United Kingdom's competitiveness in commercial clinical research (NIHR, 2025). At site level, each trust has 90 days from regulatory approval or site selection, whichever is later: 60 days to site opening and a further 30 days to recruit its first participant. This dissertation examines that 60-day set-up window.

At Lancashire Teaching Hospitals (LTHTR), Research and Innovation (R&I) records the milestone dates, reports performance against the national metric, and decides which trials to take on and when each is passed to the departments that must prepare them. Pharmacy is one of those departments, and no site opens until it has issued its green light.

Trial set-up follows different pharmacy pathways, including routine review, aseptic preparation and electronic prescribing, so the clinical trials pharmacy team's capacity for about two new commercial trials a month is not interchangeable across studies.

Demand is projected to grow by 10 to 20 per cent, increasing the importance of making the existing pathway capacity visible and usable.

Departments prepare their parts of the set-up concurrently, so pharmacy holding a 60-day goal of its own is not a problem in itself. The difficulty is that site opening waits for the last required track to finish, so a pharmacy overrun transfers directly to the site-opening date. On the available Capability and Confirmation Agreement proxy, only 3 of 24 trials fell within the 60-day allowance between 2024 and 2026; the median interval was 144 days. The tracker holds no site-confirmation date, so this later agreement stands in for it (Section 3.1).

1.2 Operational problem

Meeting the site-opening target at LTHTR cannot be addressed by pharmacy alone. Set-up capacity is fixed and distributed across general and specialist teams, while R&I decides which studies pharmacy receives and when, and therefore the volume, mix and timing of work entering the pharmacy pathway. Progressing one study consumes capacity that may be needed by another, yet the feasible number and specialist mix of studies are not visible at portfolio level. The structural problem is therefore to select the studies that LTHTR can progress within existing pharmacy capacity and identify the capacity constraint limiting further set-up. Chapter 3 describes this arrangement in full, and sets out which part of it this study models.

1.3 Research objective

The objective of this study is to determine how much of LTHTR's existing pharmacy capacity can be converted into completed set-ups of commercial medicine trials, and where effort is stranded when it cannot. Staffing is held constant throughout the analysis.

Clinical site set-up, governance and committee calendars, and staffing expansion lie outside the commissioned scope (Section 3.2). The dissertation develops a repeatable planning model that identifies which trials can be committed within the 60-day window, distinguishes individual from portfolio-level requirements, and selects the plan with the most on-time set-ups and, among tied plans, the least weighted lateness across the queue.

This dissertation makes three contributions. Empirically, it identifies the usable time remaining when work reaches pharmacy, not the nominal number of set-ups pharmacy can open, as the constraint on on-time performance. Methodologically, it combines lexicographic portfolio selection with a load-dependent shared approval queue, so on-time feasibility is assessed at plan level rather than trial level, a combination Chapter 2 identifies as a gap. Practically, it shows that selection should complement earlier notification and preparation, not substitute for time lost before the work arrives.

1.4 Research questions

Stated in terms the study can measure, the objective becomes a single critical question.

RQ. How many commercial medicine trials can LTHTR commit to pharmacy set-up within its 60-day set-up window using the capacity it already has, and what limits that number?

It is therefore approached through three sub-questions, each answered by one output of the planning model (see Table 1.1).

Table 1.1: The critical research question and its three sub-questions

Focus	**Sub-question**
Capacity ceiling	SQ1: How many commercial medicine trials, and which of them, can LTHTR commit to pharmacy set-up within the number of set-ups it can take on in a cycle?
Binding constraint	SQ2: Which constraint binds on-time completion across the observed decision points, and what changes when the cycle commitment limit is varied?
Selection rule and its robustness	SQ3: Does selecting on how many trials finish inside 60 days, with waiting time breaking ties, outperform the arrival order used today, and does any advantage survive when arrival timing, trial mix and set-up duration are varied in simulation?

1.5 Dissertation structure

The chapters follow the argument from question to answer:

- **Chapter 2** identifies the gap in the measurement, selection and queueing literature.
- **Chapter 3** states the pharmacy set-up problem and formalises it as a planning model.
- **Chapter 4** sets out the data, planning inputs, experimental design and the assumptions on which the analysis rests.
- **Chapter 5** reports what the model returns on the trust's own queues and under simulated variation.
- **Chapter 6** answers the research questions, compares the findings with published work, and states the limitations and their implications for practice and further research.

Chapter 2

Literature Review

The pharmacy clinical-trial set-up service accepts work as it arrives without explicit regard to available capacity, and some trials consequently miss the 60-day target. A planning response must identify feasible on-time commitments and limit lateness across the remaining queue. These are counted in different units, and the trust has not defined how to trade one against the other. Chapter 2 reviews the evidence relevant to both concerns and identifies the gap where they must be addressed together.

Set-up delay in UK trials is neither new nor incidental. Al-Shahi Salman et al. (2007) found a median of 44 working days for research governance approval alone across 50 NHS trusts, close to the whole of the 60-day window studied here. Hackshaw et al. (2008) recorded 52 per cent of centres taking six months or more to activate, and a decade later Duley et al. (2018) found approvals and contracts still the two most-reported inefficiencies across UKCRC trials units. Kearney et al. (2014) locate the delay more precisely: against a median 9.7 months to open a site, formal review took 16 days while the preparation preceding submission took seven. The delay therefore sits in preparation and hand-off rather than in the approval decision itself.

Two things follow for this study. Pharmacy set-up is one department inside a system with a long-documented structural delay, so the case-study pattern reported in Chapter 5 is best read as a local instance of that pattern rather than as a national prevalence estimate. And because the delay concentrates before formal review, a model of this problem must represent the time consumed before work reaches the department, not only the work the department then does.

However, before work can be prioritised, the set-up process itself must be measured, and the literature provides little support for this. Walker et al. (2018) identify 117 trial-site performance metrics, none covering set-up or activation, and the eight core metrics Whitham et al. (2018) settle on by Delphi consensus all apply after a site opens. No established method therefore exists for measuring pharmacy set-up work, and this study must define the process and its measures before modelling it.

2.1 Establishing an unrecorded process

Process mapping is the method for that step. Ofosu et al. (2021) build a map of an emergency referral system from practitioner interviews and then examine it for gaps between the intended process and the actual one. This study uses process mapping to establish the operational dependencies and information requirements that informed the model specification. It stops at that level because the trust does not record activity-level timings.

The resulting map identifies a final approval step shared across trials, which makes commitment selection consequential where that resource is constrained.

2.2 Selection under shared capacity

Which trials to start is a recognised scheduling problem. Scheduling with rejection decides which jobs to accept as well as when to schedule them, because in a heavily loaded system accepting every job increases completion delays and tardiness costs (Shabtay et al., 2013).

Slotnick and Morton (2007) give the formulation this study follows. Their model selects jobs from a pool and balances the value of accepted jobs against their weighted tardiness, and early completion is neither penalised nor rewarded, so late completion carries a cost. Lewis and Slotnick (2002) extend the decision across planning periods, allowing acceptance decisions to be updated as new jobs arrive.

Similar models have been used in clinical research, but not for pharmacy trial set-up. Deglise-Hawkinson et al. (2020) select trials for a clinical research unit that shares nursing and facility capacity. They showed that accepting an additional trial increases

waiting times for other trials, and that this effect is non-linear, which is why they treat accepting a proposed trial as a decision that requires a model.

None of these models orders its objectives. Slotnick and Morton (2007) and Lewis and Slotnick (2002) combine theirs into one financial value, because revenue and lateness penalties are both money; Deglise-Hawkinson et al. (2020) instead bound access delay and overtime. Each needs a number the trust cannot supply, a trade-off rate in the first case and a bound in the second.

2.3 Ordered objectives

This study uses pre-emptive goal programming to order its objectives. In goal programming, each objective has a target, and the model seeks to reduce the difference between the target and the result. In the pre-emptive form, objectives are placed in order of importance. A lower-priority objective is considered only if it does not reduce the result achieved for a higher-priority objective (Jones and Tamiz, 2010).

Munoz et al. (2018) show when weighted goal programming is suitable: where decision-makers can state how important one objective is against another, as their clinical institute did through expert ratings. Qi et al. (2022) weight six enrolment goals in the same way, none of them time-based. The trust has no such basis. Blake and Carter (2002) instead used the pre-emptive approach in hospital planning, fixing an economically feasible case mix before choosing among the equally feasible alternatives. Section 3.5 applies that logic.

The model in this study has two distinctive features. First, its highest-priority objective is to maximise the number of trials completed on time, which is the measure the trust is assessed on. Atsmony and Mosheiov (2023) also maximise a weighted count of jobs meeting their due dates, although theirs is a just-in-time setting in which early completion is penalised too, and they use this as their only objective. Second, the model represents a single approval step that every trial must cross, whereas the machines in Atsmony and Mosheiov (2023) are independent. The next section explains why this step limits the number of trials that can be in progress.

2.4 Modelling the shared step under uncertainty

The approval step is a queue: trials wait, and one person completes the approval. Gupta (2013) recommends the M/G/1 model for a single server whose service times follow no particular distribution, since its formula needs only the average service time and its variation, which are the only data the trust can provide. Appendix C examines the arrival assumption it makes.

Similar queues have been used to model professional sign-off. Lahiri and Seidmann (2012) model a radiologist's workstation as a single-server queue, and Svirsko et al. (2022) include waiting time that increases with workload in an optimisation model. Neither makes queue waiting time depend on which work is accepted: the first measures the queue without deciding which cases enter it, and in the second workload results from routing rather than selection decisions.

Section 3.4 brings approval waiting time into the trial-selection decision. The wait is shared by every committed trial and grows with the number committed, so a trial's on-time status depends on how many trials the plan commits, not on that trial alone.

The alternative was to simulate the full set-up pathway. Discrete-event simulation is widely used in NHS capacity planning (Demir et al., 2017; Harper et al., 2023), and Martinez et al. (2016) used it to model trial activation. The available records give coarse, retrospective segment timings, not the event-level activity, sequencing and service-time data such a simulation needs; they are sufficient for a queueing formula and not for a simulation of the full pathway.

One issue remains. The duration data are limited and highly variable, and on-time completion is a threshold outcome, so evaluating a plan on average durations does not show how often it will succeed. The selected plan is therefore tested through simulation, as Kheiri et al. (2021) test optimised NHS theatre schedules against bed-capacity breaches and Mavrotas et al. (2015) do for multi-objective models.

Two sources of uncertainty affect this test: the number of simulation runs, which sets the precision of the result (Morris et al., 2019), and the duration distribution estimated from a small sample, which Barton (2012) calls input uncertainty. Chapter 4 reports both.

2.5 Contribution of this dissertation

Table 2.1 sets the reviewed models against the problem's features.

Table 2.1: Literature review summary

	Select	**Shared step modelled**	**Value objective**	**Lateness**	**On-time count**	**Objectives**	**Setting**
Slotnick and Morton (2007)	✓		✓	✓		Weighted	Manufacturing
Lewis and Slotnick (2002)	✓		✓	✓		Weighted	Manufacturing
Blake and Carter (2002)	✓		✓			Pre-emptive	Hospital
Munoz et al. (2018)	✓		✓			Weighted	Clinical research institute
Qi et al. (2022)	✓		✓			Weighted	Clinical trials
Deglise-Hawkinson et al. (2020)	✓	✓	✓			Bounded	Clinical research unit
Kheiri et al. (2021)						Single	NHS theatres
Svirsko et al. (2022)		✓				Frontier	Hospital pharmacy
Atsmony and Mosheiov (2023)	✓				✓	Single	Parallel machines
This study	✓	✓		✓	✓	Pre-emptive	NHS pharmacy

Note. The value objective, lateness and on-time count columns record what each model optimises, not what it constrains.

Two gaps follow. First, no reviewed model combines a shared step with pre-emptive objectives: Deglise-Hawkinson et al. (2020) model a shared step but bound their other objectives rather than ordering them, and Blake and Carter (2002) order their objectives but model no shared step. This dissertation closes that gap, modelling the shared approval step as an analytical queue whose waiting time is calculated for each possible plan size, and ordering two goals pre-emptively: the on-time count first, matching the measure the trust is assessed on, and weighted lateness second. Second, no reviewed model combines an on-time count with a value objective. That gap is not addressed here, because no trial in this analysis carries a priority score (Section 3.1), and it remains open pending score coverage linked to set-up outcomes.

Chapter 3

Problem Description

3.1 The problem in practice

As Section 1.1 shows, few commercial trials met the available 60-day window. The performance problem is not explained by trial volume alone. Pharmacy completed review within 60 days in 2 of 23 trials with usable pharmacy data.

Figure 3.1 locates the pharmacy decision within the site set-up pathway and identifies the point at which the model begins and ends.

LTHTR typically brings 12–15 CTIMPs into set-up each year, counting non-commercial trials because they draw on the same teams even though only commercial ones fall under the target. The clinical trials team's service-level agreement permits up to two commercial set-ups per month. This is a nominal commitment limit rather than a measured estimate of how many trials can complete within the 60-day window.

The specialist units have no agreed capacity. Around 60% of open trials require aseptic work and therefore queue for the same small unit, which must operate below 80% utilisation under its manufacturing licence. Oncology prefers to handle one trial at a time and slows when several run concurrently. These limits do not mean that work is refused; the service reports that clinical trials are only about 2% of its weekly workload. However, the trust has not measured how many trials either unit can manage concurrently. Progress therefore depends on when the relevant team reaches a trial, rather than on whether it has overall capacity.

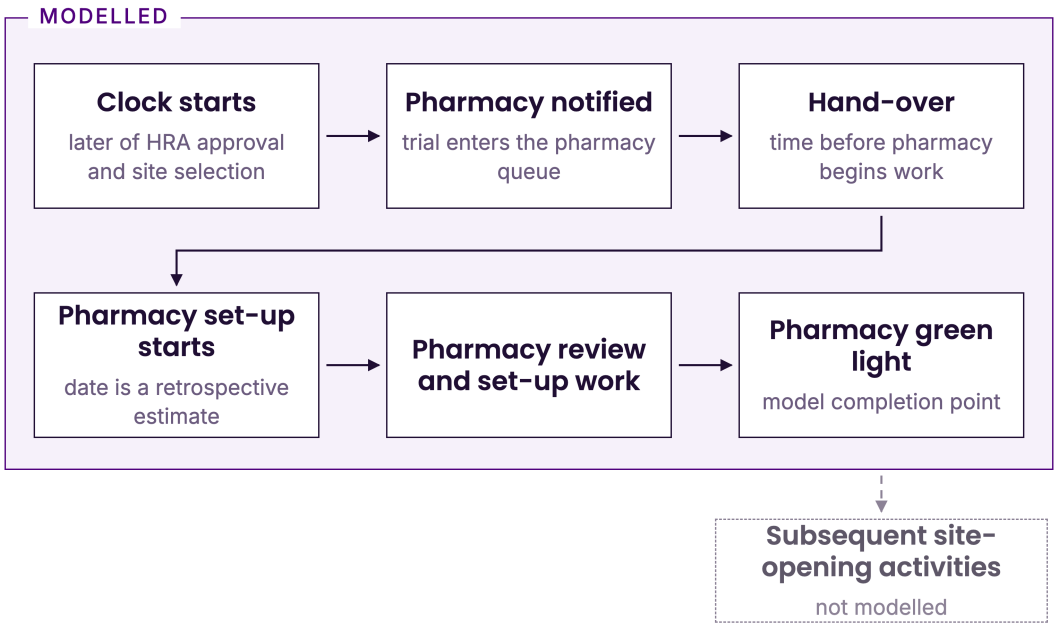


FIGURE 3.1: The pharmacy set-up sequence and the model boundary
Note. Box widths do not denote duration; Figure 4.2 shows how delays consume the 60-day window.

No formal rule allocates this scarce team time. Trials are generally taken in arrival order. Pharmacy may expedite a study it considers important outside the usual review timeline, but these decisions are informal and unrecorded. Consequently, a trial's completion date depends partly on the work that arrived before it, rather than solely on its own requirements. The model does not use the trust's Prioritisation Tool score, which ranks a study's value before set-up begins, and does not exclude trials on perceived importance. All trials within scope are therefore treated as equally worthwhile; the decision is which trials can be committed while maximising on-time completion and then minimising weighted lateness. A separate pharmacy burden rank does appear later. It records how demanding a study proved to set up, so it raises the duration floor used for the most demanding studies and marks which trials have little slack to absorb delay, not which trials are chosen or how worthwhile they are held to be. Section 4.1 gives its source and Appendix B.4 explains why it replaces the Tool's own pharmacy measure.

Two linked decisions are therefore made without a supporting method. The first is which waiting trials pharmacy starts. The second is the order in which the teams complete the work once it has started. Neither decision is recorded or reviewed, although together they shape whether pharmacy completes work within its 60-day target.

3.2 Scope and boundaries

The exclusions in Section 1.3 apply throughout, but for different reasons. Capacity is fixed: the model asks what existing staff can achieve, rather than how many staff each team should have, which is the question Grunow et al. (2004) answer by sizing a workforce to minimise total staffing cost.

Clinical site set-up is not modelled. This work, undertaken by the research delivery team, includes activities such as staff training and site initiation and is separate from the pharmacy clinical trials function modelled here. The trust records neither its start nor end date, so its duration cannot be estimated from the available data.

Governance is also excluded. The tracker shows when governance finished but not when it started, so its length cannot be worked out, and governance committees impose fixed calendar delays rather than claims on pharmacy capacity: they may block completion but do not alter the pharmacy effort a trial requires. Committee requirements are therefore recorded outside the optimisation model.

3.3 Sets, parameters and variables

Let $i \in I$ denote a trial waiting for set-up and E the eligible queue, the trials in I falling inside the scope of Section 3.2. Appendix A tabulates every symbol.

Each trial follows one of four set-up pathways, fixed by whether it needs sterile preparation and whether an electronic prescription must be built. The pathway sets the trial's set-up duration p_i in days, and records what a trial requires rather than which team supplies it.

The 60-day window is fixed by the 150-day Stage 2 definition (NIHR, 2025). It opens on the later of regulatory approval and site selection, and the site must be confirmed before it closes, so trial i carries a due day:

$$due_i = \max(\text{approval}_i, \text{selection}_i) + G, \qquad G = 60 \tag{3.1}$$

Against a planning date, δ_i is the days remaining to due_i, negative once the due day has passed.

$$w_i = 1 + a_i/D, \qquad D = 365 \tag{3.2}$$

Here a_i is the days trial i has already waited, so a trial waiting a year is twice as expensive to defer as one that has just arrived. The weight is a modelling choice, not an estimated parameter: it stops slow trials being starved without disturbing the goal ranked above it.

Three variables carry the plan. The binary commitment variable is $x_i = 1$ if trial i is committed this cycle and 0 if it is deferred. The non-negative variable d_i^+ records the days by which trial i finishes past its due day, and the binary indicator onTime_i marks a committed trial that finishes inside the window.

3.4 The approval queue

Final approval is represented as a shared, load-dependent single-server queue (Gupta, 2013). This is a conservative approximation of a process normally centred on one lead approver: delegation can shorten, but not increase, the modelled delay. Let g_n denote the expected approval wait when n trials are committed.

On-time performance is a property of the plan, not of an individual trial: approval wait grows with the number of trials committed, so the same trial may be feasible in a smaller plan and late in a larger one, and cannot be assessed until the plan is chosen. Appendix A.2 follows this through on four trials.

The formulation represents the number of trials committed by the plan explicitly. Let $z_n = 1$ if the plan commits exactly n trials and 0 otherwise:

$$\sum_n z_n = 1, \qquad \sum_{i \in E} x_i = \sum_n n\, z_n, \qquad \sum_{i \in E} x_i \le c \tag{3.3}$$

Here c is the maximum number of trials the service may commit in one cycle, so the constraints select one plan size, equate it to the number committed, and cap that number.

Let $r_{in} = 1$ if trial i would finish inside its window when n trials are committed, that is if $p_i + g_n \le \delta_i$. A trial counts as on time only when it is selected and can meet its deadline at the chosen plan size:

$$\text{onTime}_i \le x_i, \qquad \text{onTime}_i \le \sum_n r_{in}\, z_n \tag{3.4}$$

For each feasible commitment count n, the approval wait g_n and on-time indicator r_{in} are calculated before optimisation. They therefore enter the formulation as constants indexed by n, rather than as nonlinear functions of the decision variables. The selector z_n chooses one commitment count and links it to the selected trials through equation (3.3), so the resulting formulation is a mixed-integer linear programme.

3.5 The selection model

The model is a pre-emptive goal programme (Jones and Tamiz, 2010)**.** Two goals are stated in strict priority order, and a lower priority is optimised only within the value already fixed above it, because the trust has no basis for an exchange rate between them. A deferred trial receives a further planning cycle of lateness, so deferral cannot remove it from consideration, and weighted lateness is summed across the whole eligible queue rather than the committed trials alone, which prevents the model from improving its result by selecting only easy cases. Appendix A gives the goal-programming notation, the equivalence between minimising n_1^- and maximising the on-time count, and a worked instance.

The order is on-time completion first, then lateness. The second separates plans that tie on the first. Lateness is set out below because both goals are written against it.

The lateness measure. It is written as a goal rather than measured after a plan is chosen, so that deferring a trial carries an explicit cost. A committed trial finishes late by its set-up duration and queue wait, less its remaining time; a deferred trial waits a further planning cycle of L days:

$$d_i^+ \;\ge\; (p_i - \delta_i) + \sum_n g_n z_n + L\,(1 - x_i), \qquad d_i^+ \ge 0 \tag{3.5}$$

The other goal is on-time completion, targeting every eligible trial inside its window. Let n_1^- denote under-achievement of that target: the number of eligible trials not completed on time, whether deferred or committed and late. No plan can exceed $|E|$ on-time trials, so no positive deviation is required:

$$\sum_{i \in E} \text{onTime}_i + n_1^- = |E|, \qquad n_1^- \geq 0 \tag{3.6}$$

The two deviations are then minimised lexicographically:

$$\text{Lex min} \left\{ \underbrace{n_1^-}_{\text{P1}}, \; \underbrace{\sum_{i \in E} w_i \, d_i^+}_{\text{P2}} \right\} \tag{3.7}$$

P1 is the measure the trust is assessed on, the number of trials completed on time; **P2 separates plans achieving the same P1**, and because it sums across the whole eligible queue it selects the plan leaving the least lateness behind.

3.6 What the model does not decide

The model decides which trials to commit in a cycle. It does not decide the order in which those trials are worked, or which team handles each trial on a given day. It commits a set and reports what that set achieves. Ordering is handled through the queue wait g_n: the model represents the congestion that a detailed schedule would otherwise resolve, which keeps the problem linear and avoids per-team timings the trust does not record. A complementary MSc project by Inseo Lee considers sequencing after commitment; integrating portfolio selection and operational sequencing remains outside the scope of this dissertation.

Chapter 4

Methodology

4.1 Data and the methodology map

This study used a retrospective single-site operational modelling case-study design. It analysed LTHTR secondary operational records, supplemented by service clarification where required to parameterise the model, and evaluated alternative commitment portfolios across observed and simulated queues. It estimates planning performance under stated assumptions rather than causal effects of operational interventions.

The analysis combines two operational systems the trust maintains with a case-study return obtained from LTHTR operational records and service clarification. The tracker establishes the historical problem and arrival pattern, while the case-study return supplies the study-level timing, pathway and burden inputs that neither operational system provides completely; Appendix B.3 records the evidential status of each. Table 4.1 sets out what each source supplies and what limits it.

Table 4.1: The three sources and the role each plays

Source	**What it supplies**	**Role and limitation**
CTIMP 150-day tracker	Problem scale, arrivals, and the median 8-day hand-over	104 trials, 23 with a measurable pharmacy duration
Prioritisation Tool records	Prospective view of studies before pharmacy set-up	No recoverable history for trials already in set-up
The case-study return	Every per-study planning parameter	20 finished CTIMPs, August 2026

The two operational systems do not join, and the case-study return's set-up starts are retrospective estimates rather than recorded events. The Prioritisation Tool's prospective value score, set aside in Section 3.1, cannot be reconstructed for trials already in set-up, so the analyses instead use the burden rank pharmacy returned with each case study, treated as a planning-risk classification rather than an observed priority.

Two synthetic constructs supplement, but do not replace, the observed data: a synthetic standing queue, which tests the delivery reliability of a plan promising several on-time completions (Section 5.3; Appendix E.1), and generated portfolios, which test whether the selection rule's advantage survives variation in arrival timing, pathway mix and duration (Section 5.1; Appendix E.2). Neither represents LTHTR's future portfolio, and the generated cases do not test preparation or hand-over interventions.

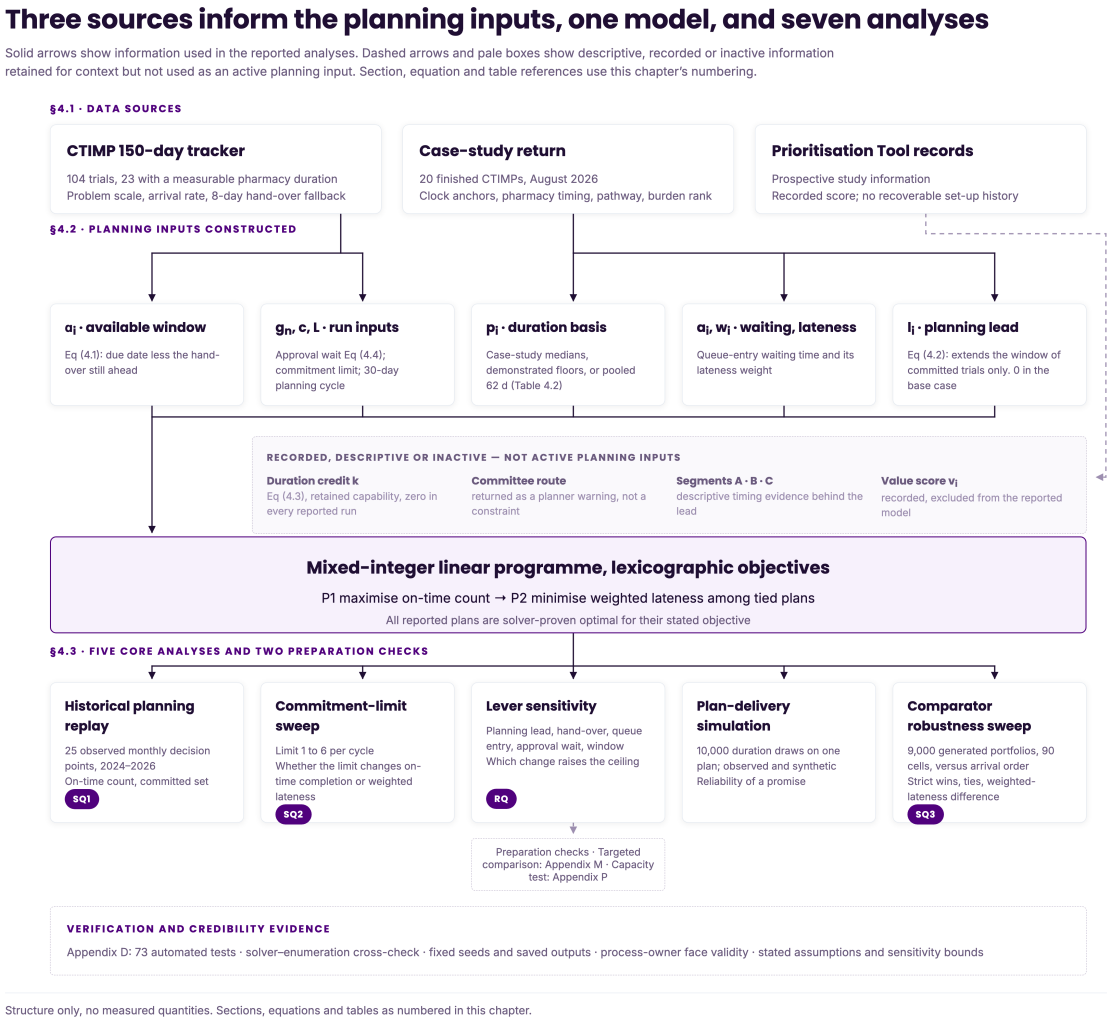


FIGURE 4.1: From operational records to research-question evidence

4.2 Constructing the planning inputs

Four quantities decide whether a trial finishes inside its window: the time available α_i, the set-up duration p_i, any preparation lead ℓ_i, and the approval wait g_n. The subsections below construct each from the available evidence. Appendix D.2 records the full parameter list, its sources and the run settings, together with the two quantities deliberately excluded.

4.2.1 Days available, which is not days remaining

A trial's available time is the period from the point at which pharmacy can begin work to its due date:

$$\alpha_i = due_i - \max(\text{today},\ \text{queue entry}_i) \tag{4.1}$$

The calculation subtracts hand-over delay that is still ahead of the trial, but not delay already served. This distinguishes the nominal 60-day deadline from the time pharmacy can actually use; Figure 4.2 illustrates the distinction.

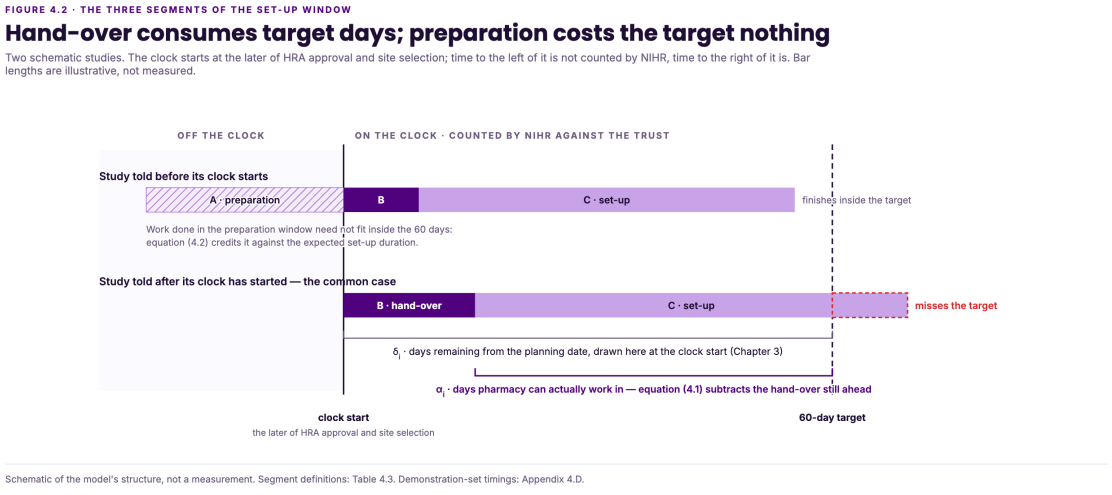


FIGURE 4.2: Hand-over consumes target days; preparation costs the target nothing
Note. A schematic of the model's structure, not a measurement. Measured distributions are in Appendix B.2.

4.2.2 Set-up duration, treated as uncertain

The duration basis differs by analysis. Current-performance regimes use elapsed case-study durations, whereas the demonstrated planning floors are derived from recorded work intervals. All duration measures end at pharmacy green light, but their start events differ; Appendix F.1 defines each interval and Appendix C considers the resulting overlap with approval wait. Across the 19 usable work intervals, the median was 79 days, with a range of 23 to 262 days; five fitted within the 60-day target. Table 4.2 summarises the five duration regimes used across the reported analyses and separates current performance from demonstrated-achievable planning conditions.

Within the demonstrated-work-floor regime, a further duration overlay applies where a trial is both on the Both pathway and classified as high burden. High burden means a pharmacy-returned rank of 5 or 6. The floors are 50 days for Routine, 30 for Prescription-only, 72 for Aseptic-only and 60 for Both, and high-burden Both studies use a 63-day overlay. Appendix F.1 distinguishes the burden-band minima, the pathway floors and the conditional overlay, and Appendix B.4 explains where the rank comes from.

Table 4.2: The five duration regimes

Regime	**What it represents**	**Where used**
0.8 times the case-study medians	Current performance 20% faster	Appendix E.2
1.0 times the case-study medians	Current performance	Appendix E.2
1.2 times the case-study medians	Current performance 20% slower	Appendix E.2
The demonstrated work floors	An achievable planning condition, with the high-burden overlay	Sections 5.1 to 5.6, Appendix E.2
The pooled first-quartile figure of 62 days	One achievable benchmark across pathways	Appendix E.2

The floors are demonstrated-achievable standards rather than forecasts or raw minima; Appendix F.1 derives and audits them. Results computed on them describe an optimistic

operating condition rather than an expected one.

4.2.3 Preparation and approval

Preparation does not shorten the total work a trial requires; it moves part of that work outside the target window. The case studies suggest that earlier starts change how much work falls within the target window rather than the observed work duration itself: work intervals did not differ detectably between early and late starters, whereas on-time outcomes did (Appendix F.2).

The reported preparation analysis represents an earlier planning lead, which for committed trials increases the time available before the 60-day deadline. It is interpreted operationally as work that can begin before the target clock starts, rather than as evidence that underlying set-up duration is shorter. Writing ℓ_i for the lead assigned to a committed trial:

$$\alpha_i^{\text{lead}} = \alpha_i + \ell_i \tag{4.2}$$

The lead is granted only to committed trials and is zero in the base case; Appendix D.2.3 gives the implementation.

The base analysis does not charge pre-window preparation against the cycle commitment limit, which isolates its time benefit from its resource cost; Appendix G.3 tests that simplification against reduced capacity.

Final approval is a shared, load-dependent queue. The wait g_n uses the Pollaczek-Khinchine formula for a single server (Gupta, 2013), exact when its assumptions hold and needing only the mean and variability of approval time, not its shape, which this study cannot observe:

$$g_n = \frac{\rho}{1-\rho} \cdot \frac{1+\kappa^2}{2} \cdot s, \qquad \rho = \min\left(0.95,\ \frac{n\,s}{A}\right) \tag{4.3}$$

Approval time is unobserved and is therefore treated as a sensitivity parameter; Appendix C documents the queueing assumptions, input proxies, and potential overlap with the observed set-up duration.

4.3 Experimental design

Every run builds a plan by the process in Figure 4.3, differing only in what it varies before that process starts and records after it ends. The model maximises on-time completions first and then minimises weighted lateness among tied plans. For Section 5.6, the model also reports the least-late plan at each achievable on-time count to display the trade-off; these are count-constrained comparisons, not alternative lexicographic optima. Appendix D.2 provides the formulation and solver settings.

Table 4.3: Analysis-to-question map

Analysis	**Data basis**	**What varies**	**What it measures**	**Question**
Historical planning replay	Observed queues	25 monthly decision points, 2024 to 2026	On-time trials, committed set	SQ1
Commitment-limit sweep	Observed queues	Commitments per cycle, 1 to 6	On-time count and weighted lateness	SQ2
Lever sensitivity	Observed queues	Preparation, hand-over, queue entry, approval, target window	Which change raises the ceiling	RQ, SQ2
Plan-delivery simulation	Observed and synthetic queues	10,000 duration draws on one plan	Reliability of a promised outcome	RQ
Comparator robustness sweep	Generated portfolios	9,000 portfolios, 90 scenario cells	Strict wins over arrival order, and their cost	SQ3

Three different denominators apply, because the inclusion rule differs. Nineteen of the twenty case studies enter the planning replay, being those satisfying the strict anchor rule of holding both an HRA approval and a site-selected date; BALLOON is excluded for having no site-selected date. The work-interval measures instead use the nineteen with a recorded set-up start, excluding BRISOTE; the pharmacy-held interval uses all twenty. Appendix D.2 holds the run settings, of which one hands-on day per approval is the only input in Chapter 5 not measured.

Two simulations are reported separately. The plan-delivery simulation tests the reliability of a selected plan under duration uncertainty, whereas the robustness sweep

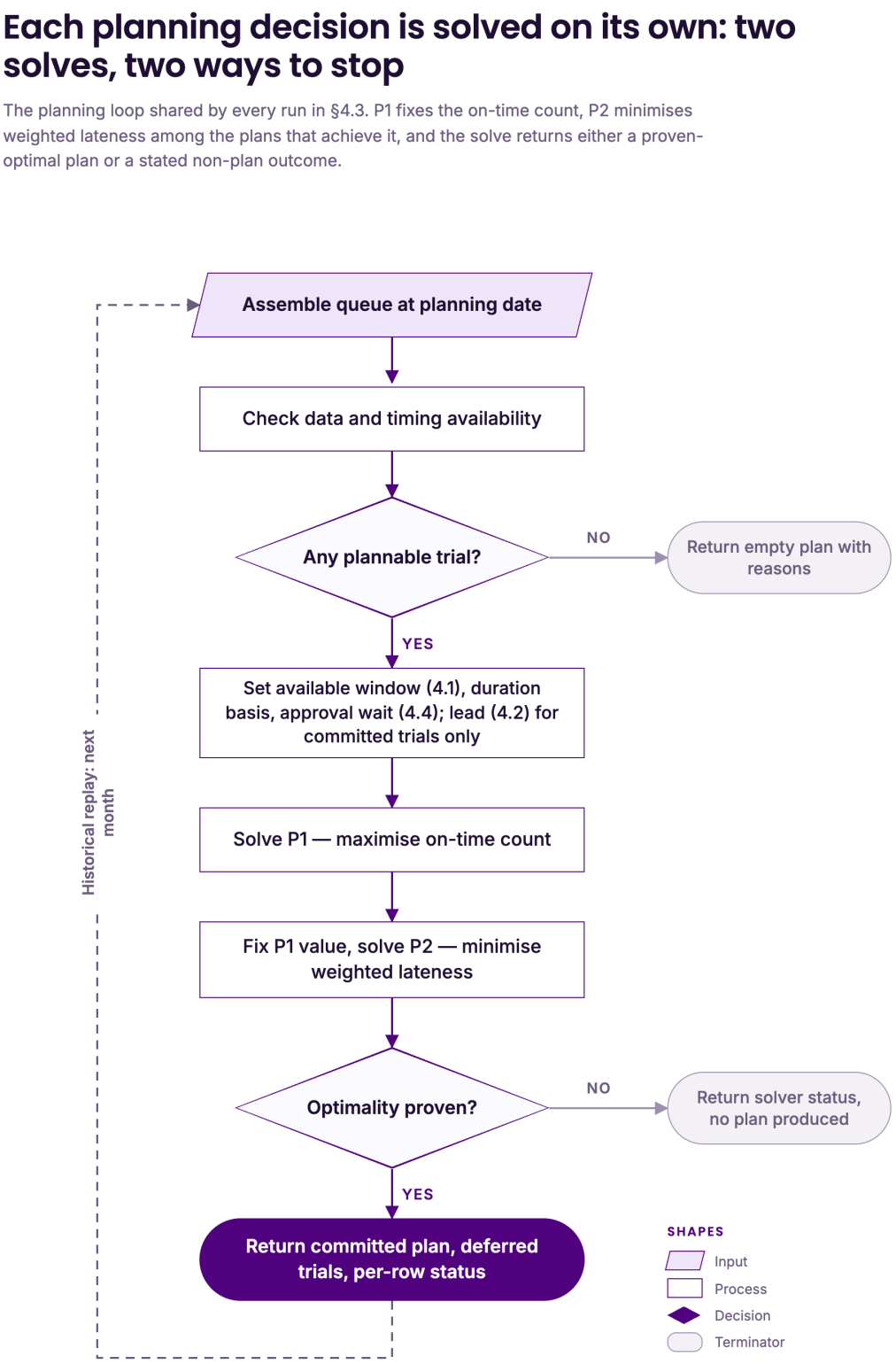


FIGURE 4.3: Solution construction process.

Note. The dashed return arrow applies to the historical replay only: each month is solved independently and no slot is held open for a later one.

tests whether the selection rule outperforms arrival order across generated portfolios. Appendix D.2 specifies their distributions, settings and reporting rules.

4.3.1 The comparator and the robustness sweep

The comparator is arrival order (Section 3.1): it commits the longest-waiting eligible trials up to the same commitment limit as the model. Both plans run through Figure 4.3 under identical windows, duration assumptions and approval waits, so any difference arises from the selected set alone. Because practice may informally expedite some studies, the comparator result is an upper bound on the model's advantage over routine practice.

The sweep generates portfolios that vary arrival timing, pathway composition and duration regime, and reports the count in which the model places strictly more trials inside the window, with ties and the weighted-lateness difference beside it. Appendix D.2.6.1 specifies the design and states why ties are read at cell level rather than pooled. **Table E.3 summarises the sweep by duration regime and is the principal test of SQ3**; Appendix E.2 holds the grid.

4.4 Credibility, assumptions and boundaries

Ethics and data governance. The University Ethics Decision Tool classified this as secondary-data work not requiring ethical approval, confirmed with the supervisor. LTHTR authorised reporting of trial identifiers; raw operational extracts and clarification returns are withheld to protect confidential service information.

Verification, reproducibility and face validity. Seventy-three automated tests verify input construction, optimisation logic, simulation behaviour and agreement between enumerated and solver solutions, with the full inventory in Appendix D.1. All simulation runs use fixed seeds and save dated outputs, so Chapter 5 figures can be regenerated exactly. R&I and pharmacy reviewed the process representation and input definitions, which constitutes face validity from process owners rather than independent validation.

A known overlap, bounded rather than removed. Each set-up duration ends at the pharmacy green light, which is the approving signature, so it already includes part of the wait that equation (4.3) adds separately. That component is dated in no system and

cannot be separated, so the overlap is retained as a bounded conservative approximation rather than an empirically identified separation of work and approval wait; Appendix C gives the assumption audit and the sensitivity result.

What the evidence cannot establish. Three limits bound every result. Set-up start dates are estimates, so the durations of Section 4.2.2 carry recall error the data cannot quantify. Hands-on approval time is unmeasured, so the checks above verify the implementation and not the formulation behind equation (4.3); testing that needs a separate model of the approval-signature queue, recorded as future work in Chapter 6. And the generated data of Section 4.3 answer questions the observed queues cannot, but do not evidence the trust's own portfolio. All three are handled by stated assumption, sensitivity analysis and simulation rather than presented as observed fact.

Chapter 5

Results

This chapter reports what the planning model returns on the trust's own queues. It gives the capacity ceiling and which trials reach it, tests how reliably a plan delivers what it promises, and identifies what limits the number and what raising it costs. Table 4.3 maps each analysis to the sub-question it answers.

5.1 The model's decision on the real queues

The model was solved at each of the 25 decision points of Section 4.3. Figure 5.1 sets its plan against what the service actually did.

Four distinct trials are ever feasible for on-time completion across the twenty-five decision points, and they are exactly the four that met the target in reality. The figure is an upper bound on rescueable trials over the whole period, not a count per cycle. The model identifies these trials in advance rather than in hindsight; it does not uncover any additional trial that the service could have brought in on time.

Compared with the commit-the-longest-waiter comparator, which selects the longest-waiting eligible trials first, the optimisation model produced at least one on-time completion at 9 of the 25 monthly decision points at each of the one- and two-set-up limits, compared with 6 decision points under the comparator. At commitment limits of three or more, both approaches produced at least one on-time completion at the same 9 decision points and selected the same trials. The 9 months cover only four trials. Because the plan is remade each month, a trial that still fits is re-committed and counted again. The comparator is a stated arrival-order rule, not an observed record of practice.

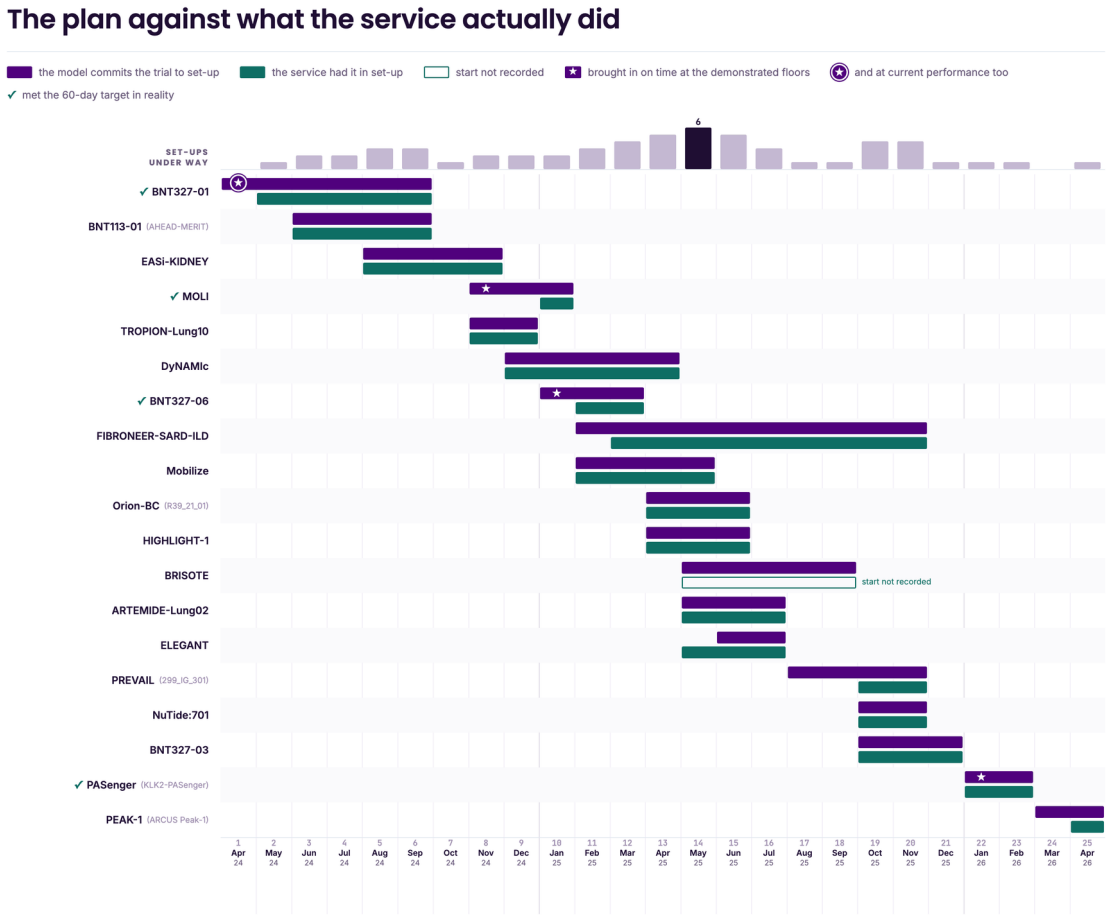


FIGURE 5.1: The plan against what the service actually did, by month of commitment rather than by date

A sweep of 9,000 synthetic portfolios tests whether that pattern holds beyond the trust's own queues. At current performance the model never beats the arrival-order rule for on-time completions. At the demonstrated floors it wins in 36% of cases, each win costing 24.3 weighted days of lateness elsewhere, and it helps most when trials arrive early and the aseptic share is low. Appendix E.2 holds the design, the limitations and the full grid.

5.2 Why the model picks these and not others

The model starts the right trials sooner. Three of the four trials it brings in on time are among the six it starts earliest: MOLI by 34 days, BNT327-06 by 22 days, and BNT327-01 by 17 days. Appendix H sets the model's start date against the service's own for all nineteen.

The service's own head start bought little: of the twelve trials it began before the model would, eleven missed the target anyway. Waiting is what consumes the window, and the trial that can still finish is usually the one that has just arrived; Appendix H works through November 2024.

Figure 5.2 shows why so few can finish at all, placing the 19 studies with a computable clock start on one axis measured in days from that start.

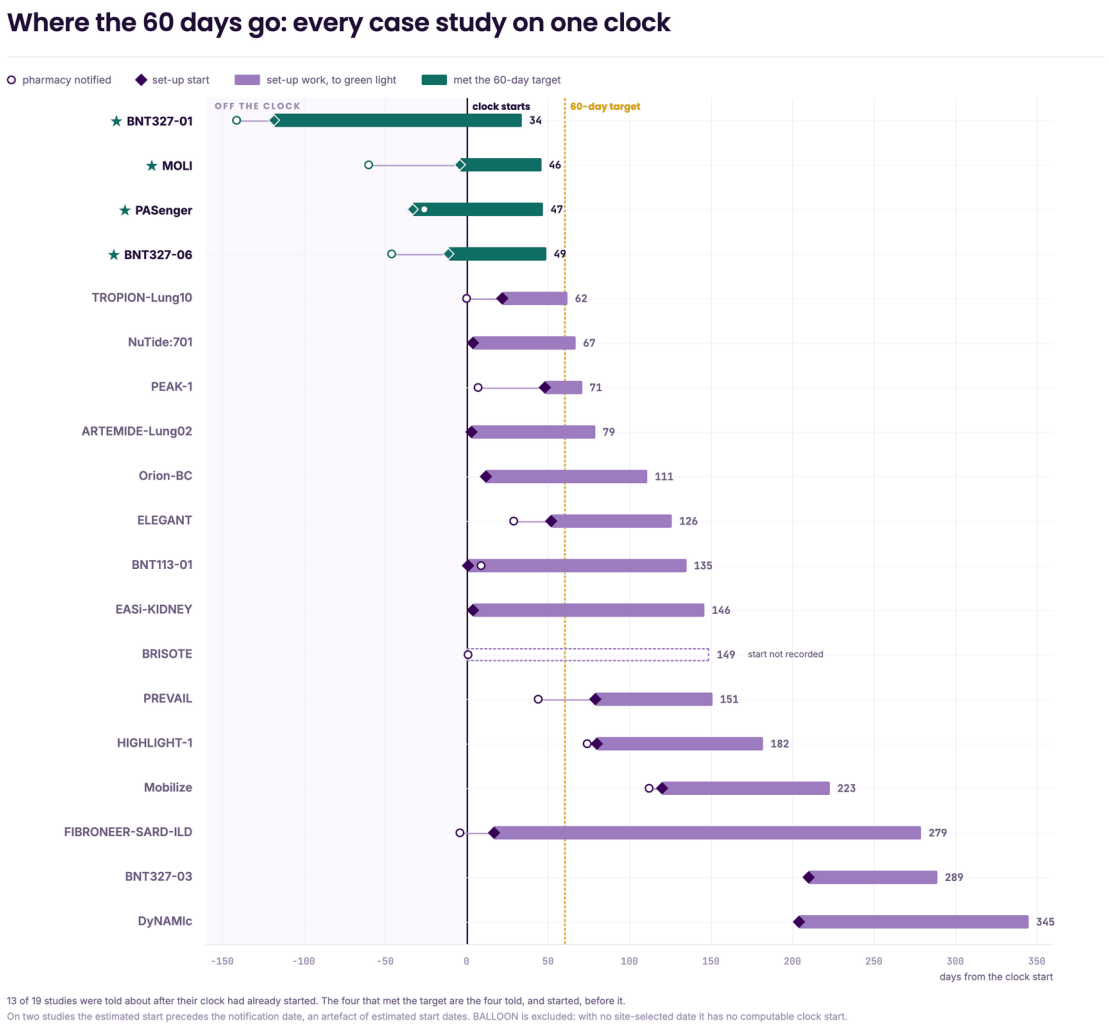


FIGURE 5.2: Where the 60 days go: every case study on one clock

Four trials met the target, and all entered the pharmacy queue before the 60-day clock began.

The observed work intervals did not differ significantly between early and late starters ($p = 0.71$). Given the small case-study base, this result does not establish that earlier preparation has no operational effect. Instead, earlier starts moved some work outside

Table 5.1: The four trials that met the 60-day target

Trial	**Days entered before clock start**	**Work interval (days)**	**Clock interval (days)**
MOLI	4	50	46
BNT327-06	11	60	49
PASenger	33	80	47
BNT327-01	118	152	34

the 60-day window. Two of the four trials met the target despite requiring more than 60 days of work because their set-up began before the window opened. The window did not stretch; the work began earlier. No recorded trial characteristic predicted an early start (Appendix F.2).

Where the plan falls short, it does so by relatively little. Four trials missed the target by 20 days or less in reality and, under the plan, fall 10 to 29 days short of their available windows. This shortfall reflects insufficient time in the window, not a failure to choose between trials. No selection rule can recover those days; Section 5.4 examines what would add them.

5.3 Plan delivery under duration uncertainty

A solved plan produces one on-time count from a single planned duration for each pathway, so the uncertainty run of Section 4.3 and Appendix D.2.6.2 replays each committed plan 10,000 times to report how often that count is actually met. Appendix E.3 reports what the real February queue delivers, and Figure 5.3 the synthetic one.

In the real February queue, neither duration basis produced a plan promising an on-time completion. The simulation therefore describes outcome variability rather than delivery reliability for a positive promise; approval wait is held at its expected value throughout, so every reported figure understates total uncertainty. Appendix E.3 reports the distribution.

The synthetic standing queue of 16 trials described in Appendix E.1 is a scenario construct, not a forecast of LTHTR's portfolio. At current performance it promises and delivers zero in all 10,000 runs. At the demonstrated floors it promises three, and Figure 5.3 shows the simulated outcomes: three in 32.6% of runs, two in 48.5%. The

deterministic optimum is therefore unsuitable as a service commitment: the promised three lands in about a third of runs, two is the most likely outcome, and one is the highest count reached in at least 90% of runs.

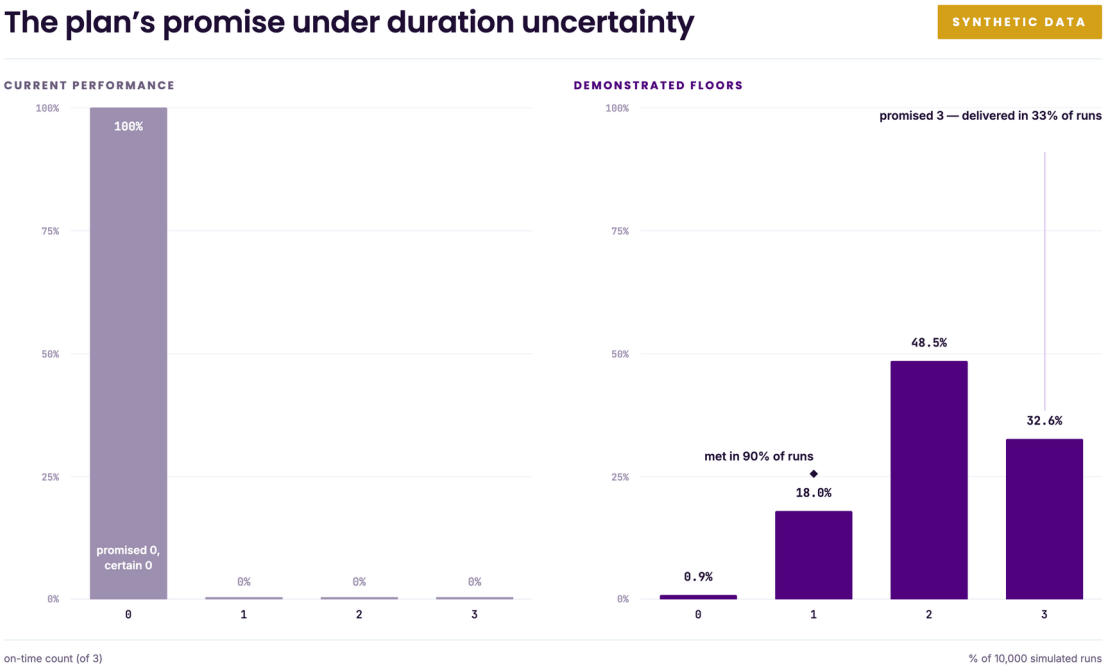


FIGURE 5.3: The plan's promise under duration uncertainty (synthetic queue)

Where a deterministic margin is smaller than the observed spread of set-up durations, the result is a tie-break rather than a finding: a one-day lateness label carries no information, and a trivial change of input reverses it. Appendix E.3 works through the case behind this.

5.4 What would have to change

Six settings could change the results. Figure 5.4 compares them on the two outcomes, and the two rankings disagree in a way that matters. Table G.2, in Appendix G.2, gives the tested range and the values behind each point, all at the demonstrated floors under the high-burden overlay.

The levers affect the two outcomes differently. Removing the hand-over gap produces the largest reduction in weighted lateness but few additional on-time trials, because many affected cases remain unable to meet their available windows. Reducing the pharmacy

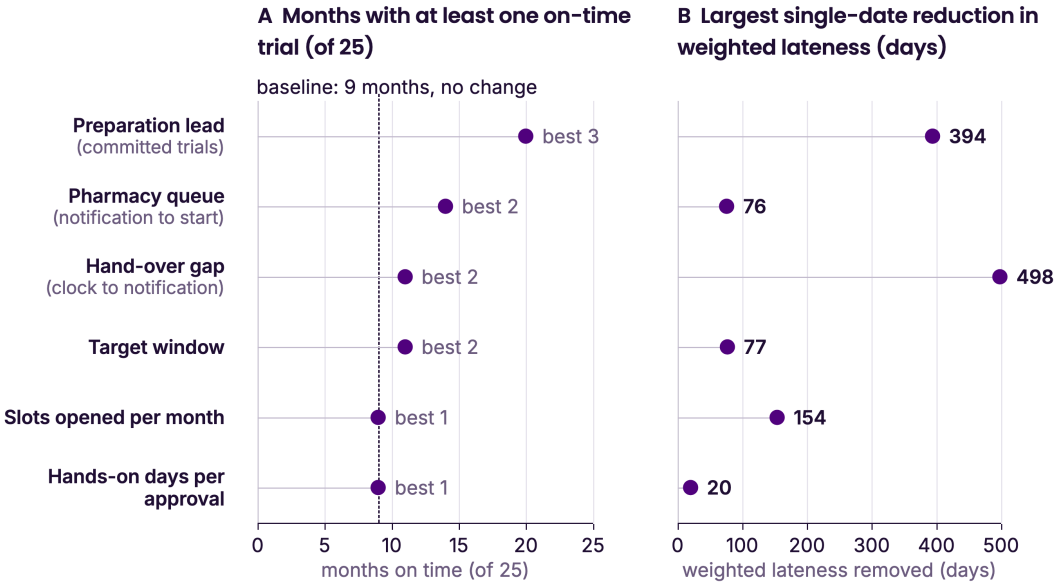


FIGURE 5.4: How the six settings compare on the two outcomes

queue removes less lateness but creates more on-time completions by affecting trials near the threshold. Appendix G.2 gives the values.

Apart from preparation, no setting reaches three on-time trials in any month: the two delays and the target window reach two, and slots and approval time never pass one. Everything except preparation therefore moves more months onto the target without raising the ceiling. Appendix G.2 draws four of them as surfaces.

Preparation is the only setting that creates this opportunity, and Figure 5.5 shows its effect on the portfolio. Appendix G.3 tests the capacity assumption of Section 4.2.3 by reducing the concurrent capacity available to the plan.

Approval time is the only one of these settings without direct measurement, because the August 2026 questionnaire gave a range rather than a value. The commitment limit is a maximum, not a requirement: if opening six set-ups would create a long signature queue, the model commits fewer trials instead. Approval time therefore limits how much capacity is usable, not the on-time ceiling. Appendix G.2 reports the wait at each level and how many set-ups are worth opening.

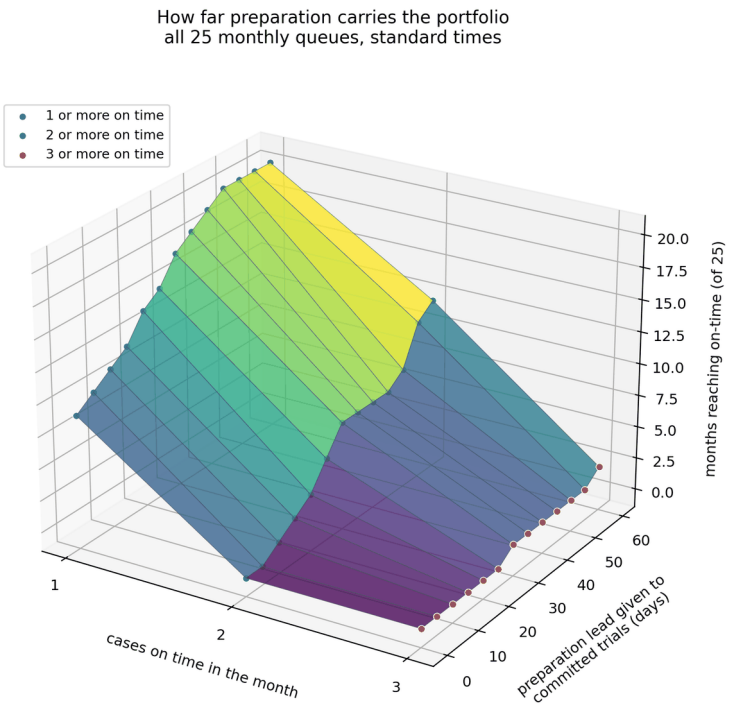


FIGURE 5.5: How far preparation carries the portfolio

5.5 Which studies preparation should reach

Figure 5.4 makes preparation the strongest lever; pharmacy burden says which studies should receive it.

The three burden bands showed no detectable difference in work duration (Kruskal-Wallis $p = 0.557$), though groups this small would not reveal a moderate one. What burden does change is the fastest observed duration in each band: 23 days for low burden, 40 for medium burden, and 60 for high burden (Table F.3). This leaves 37 days of slack within the 60-day target for low-burden trials, 20 for medium-burden trials, and none for high-burden trials. Burden is therefore used as a planning-risk classification built on that observed floor, not as a demonstrated cause of longer set-up.

The median time between the target clock starting and pharmacy receiving the study is 14 days (Appendix B.2); low- and medium-burden trials can absorb this delay. High-burden trials cannot absorb even one day; Figure 5.6 sets the three bands against the window. The exclusion is therefore structural. Of the six high-burden trials, the two that began before the clock started met the target, whereas all four that began afterwards

missed it (Fisher's exact test, $p = 0.067$). The pattern fits the proposed mechanism but does not establish it.

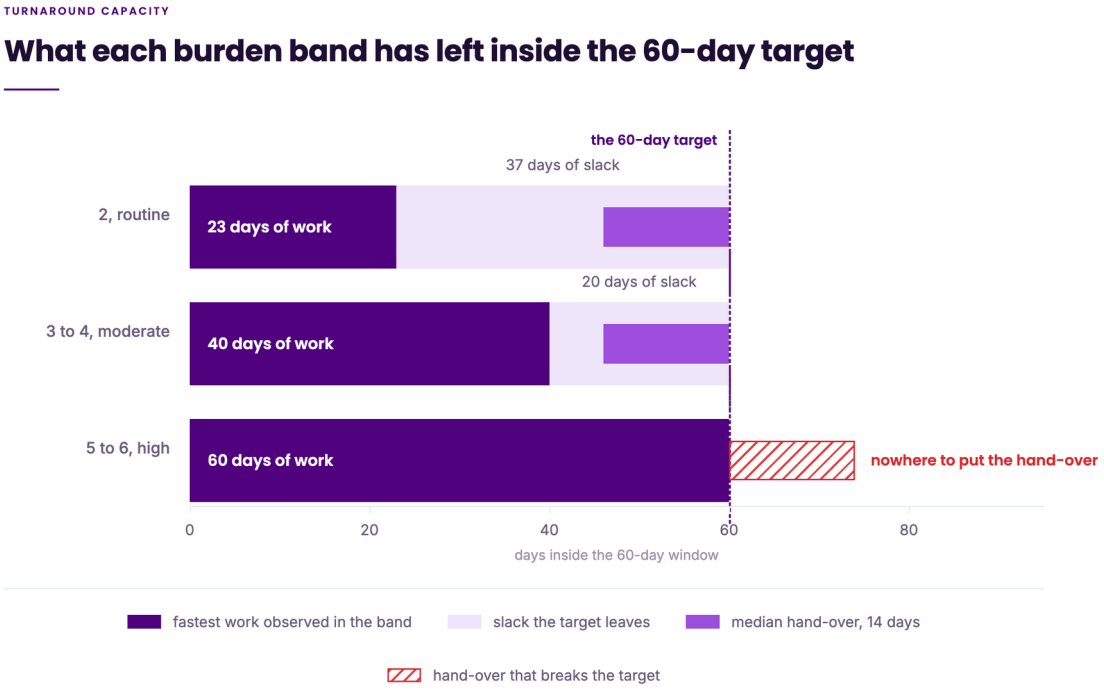


FIGURE 5.6: What each burden band has left inside the 60-day target

Preparation is the only lever these studies have, and it is not aimed at them: burden rank does not predict which trials start early (Fisher $p = 1.000$, Appendix F.2). All six high-burden trials are aseptic, and eleven trials in the set are, so the aseptic flag identifies every one of them at notification and five others besides.

Targeting preparation at the aseptic-flagged studies concentrates effort on identifiable high-burden risk for about half the earlier notification, but its achievable on-time count is lower than preparing every committed trial, because it leaves the rescuable non-flagged trials untreated (Appendix G.1).

The lead must be sized against the variation in durations rather than against the shortfall it closes. Appendix E.3 shows the four-day fix for BNT327-06 delivering half the time. Reading a reliability as the proportion of runs in which a plan delivers the stated outcome, a high-burden study planned at the 63-day floor is 50 per cent reliable with no lead, 75 per cent with a 28-day lead, and 90 per cent only at a 63-day lead, a second window on top of the first. Preparation is what makes a high-burden study possible; a few days of it buy a plan that says yes rather than a delivery that can be relied on.

5.6 The trade-off the service faces

Preparation is the intervention considered here. This section reports the trade-offs it creates because the model ranks two objectives that do not always align.

The trade-off between on-time completion and weighted lateness appears only where preparation remains insufficient. On the largest queue, 1 May 2025, a 21-day lead makes each rescue cost additional lateness, whereas a 35-day lead brings two trials in on time at the lowest weighted lateness of any plan; on the 1 February 2025 queue the two objectives never conflict at any lead. Figure 5.7 sets the two leads side by side and Appendix G.2 reports the frontier values.

The conflict between finishing on time and minimising lateness is itself a symptom of insufficient preparation. A governance rule that ranks lateness first gives up nothing when preparation is adequate and two rescues when it is not, so it is the wrong rule to fix in advance.

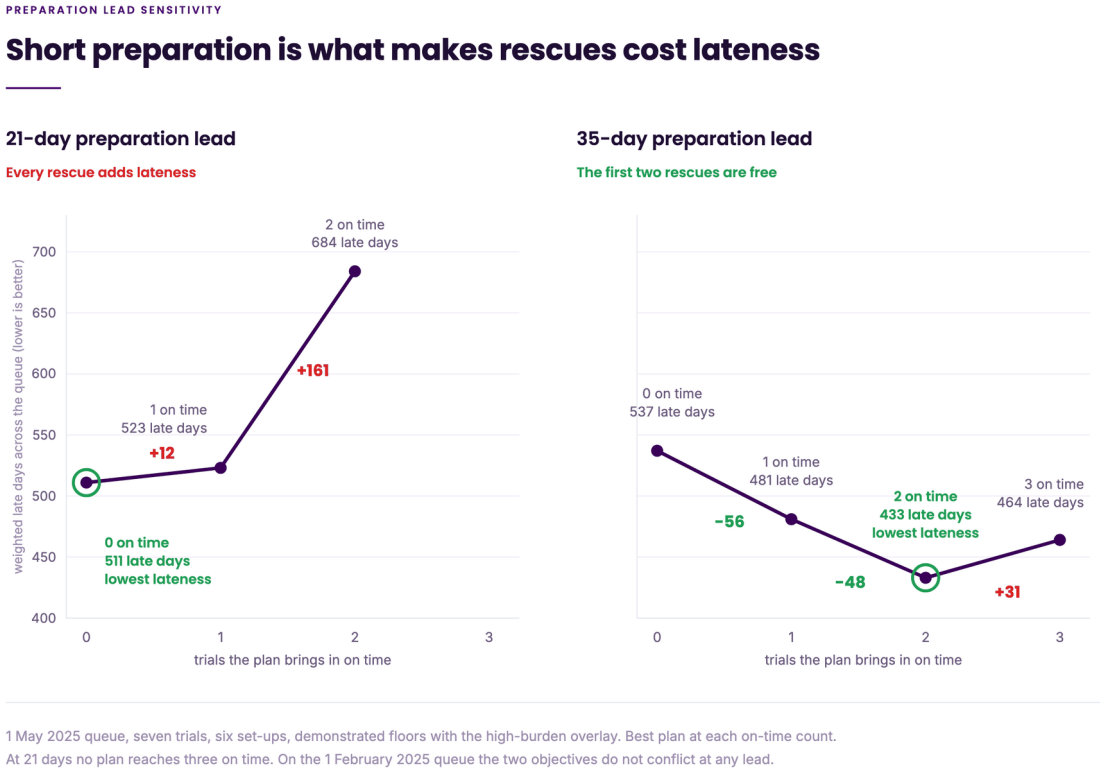


FIGURE 5.7: What a rescue costs at two preparation leads

The second trade-off is between the on-time count and the preparation it requires, the cost the trust pays in days R&I must find. Figure 5.8 reports the least preparation that first reaches each count.

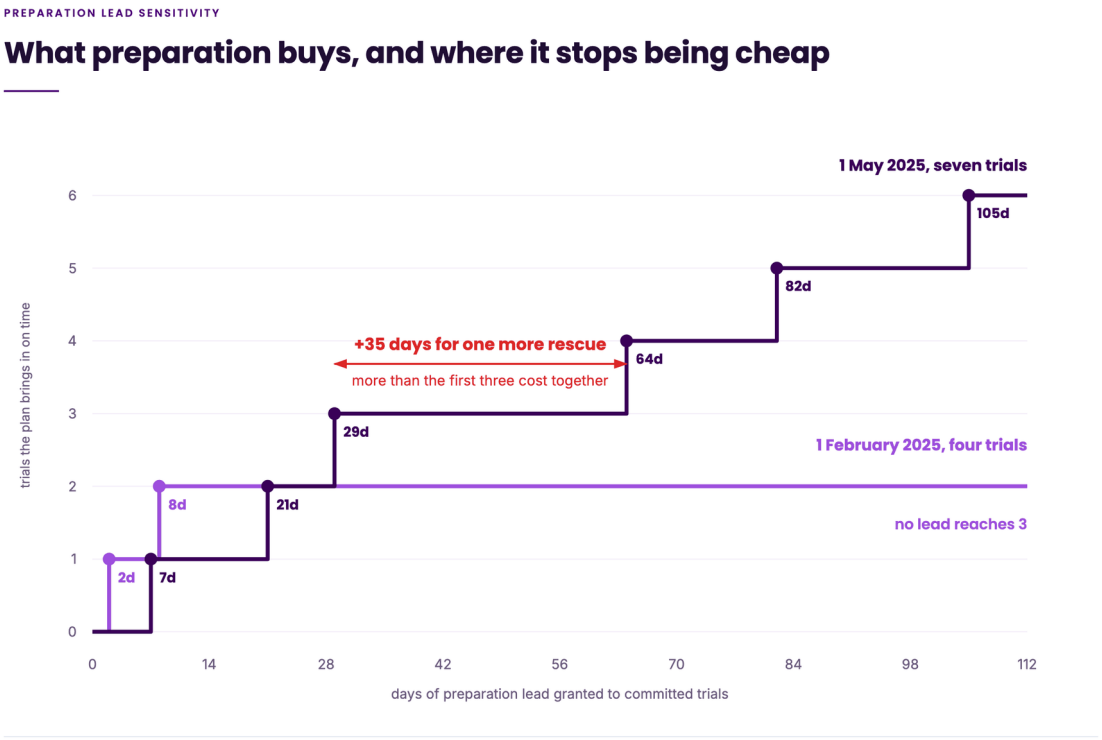


FIGURE 5.8: What preparation buys, and where it stops being cheap

Two things follow.

First, capacity never limits the on-time count. Each rescue requires one set-up slot, so a plan that rescues three trials uses only three of the six available slots.

Second, what a queue can reach is set by its own composition. A lead only repays a shortfall the queue already carried, so it buys a rescue cheaply while a trial is near its window and buys little once a trial is far past it. Across the whole range Figure 5.8 covers, the February queue never passes two. That ceiling belongs to the queue rather than to any lever, and it is the honest limit on what scheduling can promise.

Capacity does not stop working above the rescue count, however. Figure 5.9 holds the preparation lead at 35 days and varies the set-ups opened per month against the on-time count. Lateness falls with every slot opened at each on-time count, while the pharmacy burden a plan commits rises with both. Opening more set-ups is therefore a lever on

how late the rest of the queue runs rather than on how many trials meet the target. Appendix G.2 gives the values and the derived burden scale.

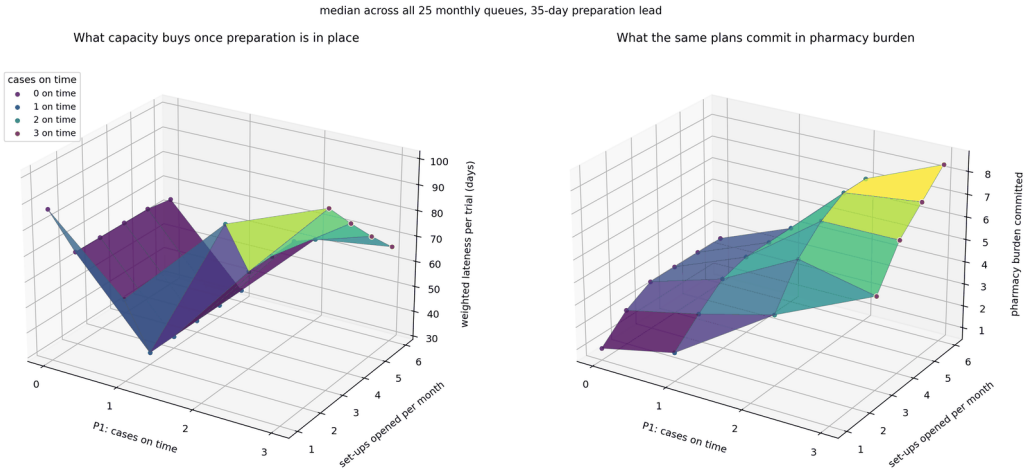


FIGURE 5.9: What capacity buys, in days and in burden

Chapter 6

Conclusion

6.1 Answers to the research questions

Table 6.1 answers the critical question and its three sub-questions, and cites the evidence behind each; their full wording is in Section 1.4 and Table 1.1.

Table 6.1: Answers to the critical question and sub-questions SQ1 to SQ3

	Answer
RQ	Four distinct trials were ever feasible for on-time completion across the 25 retrospective decision points. Time remaining when a trial reaches pharmacy, rather than nominal set-up capacity, limits this number. Preparation is the only tested lever that raises the ceiling, because it creates time before the clock starts (Sections 5.2 to 5.6; Table G.2).
SQ1	Across the 25 retrospective decision points, four distinct trials could be committed and completed on time within the cycle commitment limit. This is an observed upper bound under the recorded timing conditions, not a monthly throughput estimate. Table 5.1 identifies the trials (Sections 5.1 to 5.2).
SQ2	Usable time remaining when a trial reaches pharmacy binds on-time completion. Neither the cycle commitment limit nor approval queue binds: raising the limit adds no on-time completion across 25 decision points, and the model leaves slots unused when commitments add approval wait without improving performance. Approval time determines usable capacity, not the on-time ceiling (Sections 5.4 to 5.6; Appendix G.2).

continued overleaf

	Answer
SQ3	The optimisation model improves on the arrival-order comparator only when capacity is tight: at one or two set-ups per month it selects a better committed portfolio, whereas from three onwards both approaches select the same trials. Its advantage is not robust across simulated portfolios: it appears only at the demonstrated duration floors, where the model is better in 36 per cent of runs and equal in the remainder by construction (Section 5.1; Appendix E.2).

6.2 Contribution and comparison with published work

The contribution is a calibrated, transparent translation of LTHTR's own pharmacy set-up process, its capacity, pathways and shared approval queue into a portfolio-level planning model. The repeatable model makes commitment decisions explicit, compares feasible portfolios before work is committed, and distinguishes nominal from usable capacity. Its wider analytical implication is limited but clear: before changing downstream capacity or selection rules, a service should establish where the target window is consumed and whether enough usable time remains when work reaches the focal team.

The study also provides a visual mock-up of a future planning interface, translating the model into a user-facing workflow (Appendix I); it is a design artefact, not a deployed or validated system.

The findings refine the practical implication of the selection literature: selection can improve the allocation of commitments and reduce lateness across the queue, but it cannot recover time lost before pharmacy receives a trial. Earlier notification and preparation therefore create the conditions in which selection can improve on-time performance (Slotnick and Morton, 2007; Munoz et al., 2018; Qi et al., 2022; Sections 5.1 to 5.6). The contrast with Deglise-Hawkinson et al. (2020) is instructive. They show that accepting an additional trial raises waiting non-linearly, which is what makes acceptance worth modelling. The same mechanism operates here but does not bind: raising the commitment limit changed the on-time count at none of the 25 decision points, because most of the window was spent before pharmacy received the work. Diagnosing this locally requires the model boundary to include delays that consume the target window before the focal department receives the work; otherwise, it may attribute poor on-time performance to a downstream resource that cannot determine it. Pharmacy is the first module of a wider trial set-up portfolio; comparable, consistently defined process, duration and

capacity data from other departments would allow interdependent workstreams, delay locations and feasible improvement levers to be assessed jointly.

6.3 Limitations

The comparison with arrival order is limited by the comparator, not by the optimisation result. The comparator is a stated rule, not a record of routine practice, which already expedites some important studies. Any reported advantage is therefore an upper bound on the benefit over current practice, and the limitation bears on SQ3 alone. Similarly, agreement between the model and the four trials that achieved the target is retrospective consistency rather than prospective validation (Section 5.1; Appendix E.2).

The four-trial upper bound is a conditional planning estimate, with its fourth on-time place particularly sensitive to the high-burden duration assumption (Section 5.3). The analysis rests on a small case-study base, estimated timing data, uncertain set-up durations and an assumed approval-time input. Sensitivity and simulation analyses test these uncertainties, but cannot independently validate the capacity ceiling or the reliability of the preparation result. In particular, holding approval wait at its expected value understates total operational uncertainty; the estimate should therefore not be assumed to transfer unchanged beyond the study setting (Sections 4.2 to 4.3, 5.3–5.4; Appendices E.2 and G.2).

Preparation is not capacity-free: the base model does not charge pre-window work against the commitment limit, so it may overstate its benefit. The sensitivity analysis in Appendix G.3 shows that preparation still increases the number of decision points achieving at least one on-time completion across the tested capacity range, although the three-trial result disappears at two set-ups.

Finally, the model selects trials for commitment but does not sequence work after commitment. The evidence therefore concerns portfolio selection only and does not establish whether a sequencing intervention could improve performance (Section 3.6).

6.4 Implications for practice and for further research

The results imply four immediate actions for LTHTR: record hands-on approval time to calibrate the approval queue; flag aseptic studies at notification for targeted early

preparation; plan to a reliable lower outcome rather than the deterministic maximum; and use selection to support, rather than substitute for, earlier preparation and handover. Together, these actions make commitment decisions more transparent (Sections 5.1 to 5.5). Section I.7 sets these actions out as an implementation pathway.

Four research directions follow from the limitations above:

- Extend the model to integrated selection and sequencing, which the present analysis cannot address.
- Measure pathway-specific work, approval service times and rework, so that resource use can be modelled in staff time rather than commitment counts.
- Optimise early preparation under finite capacity, including the activity it displaces.
- Prospectively evaluate the model's recommendations against observed practice, under demand growth, changing pathway mix and interdependent cross-service workflows.

Appendix A

Notation and Worked Instance

A.1 Notation

Table A.1: Notation used in the selection model of Chapter 3

Symbol	Meaning	Units
$i \in I$, E	Trial awaiting pharmacy set-up; the eligible queue	n/a
G, due_i	Set-up goal (60), due day $\max(\text{approval}_i, \text{selection}_i) + G$	days
p_i	Set-up duration of trial i, from its pathway	days
δ_i	Days remaining to due_i at the planning date, negative once breached	days
a_i, D	Days trial i has waited; weight divisor (365)	days
w_i	Claim weight, $1 + a_i/D$	dimensionless
c	Set-ups the service can commit in a cycle	trials
L	Planning cycle length (30)	days
g_n	Expected approval wait when n trials are committed	days
r_{in}	1 if trial i finishes inside its window at committed count n	binary
x_i	1 if trial i is committed	binary
z_n	1 if the plan commits exactly n trials	binary
onTime_i	1 if trial i is committed and lands inside its window	binary
d_i^+	Days by which trial i finishes past due_i	days
n_1^-	Eligible trials not completed on time; under-achievement of the P1 target	trials

A.2 Worked instance: why on-time is a property of the plan

This instance shows the mechanism Section 3.4 describes: the same trial can finish inside its window in a small plan and outside it in a large one, so committing more trials can leave fewer of them on time.

Durations here are illustrative, at 36 days. The measured durations cannot be used for this purpose. Pooled across the tracker they run to about 95 days against a 60-day window, which puts every trial outside the target whatever the plan, and removes the question the instance is meant to show. The distinction the instance makes is therefore about the model's behaviour, not about LTHTR's current position.

The expected approval wait, computed at the model's default settings, rises slowly and then sharply:

Table A.2: Expected approval wait g_n at each committed count

Committed count n	1	2	3	4	5	6	7	8	9
Wait g_n, days	0.20	0.46	0.79	1.25	1.91	2.96	4.86	9.39	32.32

Four trials from the eligible queue, each needing 36 days of set-up, differ only in how much of their window remains:

Table A.3: The four trials of the worked instance

Trial	Days remaining δ_i	Slack over the 36-day set-up
A	50	14.0
B	44	8.0
C	38	2.0
D	12	none, already short

A trial finishes inside its window when $\delta_i \geq p_i + g_n$. Applying that at each plan size gives the rescue matrix r_{in} the model reads:

Three things follow, and each answers something the chapter asserts.

Rescueability is not a property of the trial. C survives at five committed and fails at six. Nothing about C changed; the queue it joined grew. This is why the model cannot decide trial by trial before the plan is chosen, and why the load selector z_n of equation (3.3) exists.

TABLE A.4: The rescue matrix r_{in} at each plan size

Plan size n	1	2	3	4	5	6	7	8	9
A rescueable	yes	yes	yes	yes	yes	yes	yes	yes	no
B rescueable	yes	yes	yes	yes	yes	yes	yes	no	no
C rescueable	yes	yes	yes	yes	yes	no	no	no	no
D rescueable	no	no	no	no	no	no	no	no	no

Raising the commitment limit eventually makes performance worse. Counting trials on time as the plan grows gives three at five committed, two at six and seven, one at eight, and none at nine. The on-time count peaks and then falls, because past the peak the queue costs more days than the tightest rescueable trial has left. That is the answer SQ2 asks for, and it is why "take every slot you can get" is the wrong reading of a capacity case.

A trial can be beyond help. D has 12 days against a 36-day set-up and is outside its window at every plan size. It contributes nothing to P1 whatever the model does, which is why P2 exists: deferring D still lets its lateness grow, so the model is not free to ignore it.

Trial labels here are illustrative. The waits are computed from `Experiments/optimiser/gate.py` at default settings; the durations and remaining days are chosen to make the mechanism visible.

Appendix B

Data Sources, Cleaning and Operational Inputs

B.1 The Tracker, its Structure and its Cleaning

Source, unmodified: `CTIMP Trials 150 Day Metric Tracker- 2026-07-30 SE.xlsx`. Outputs: a cleaned table holding one row per trial, a model-ready extract of the fields the model uses, and the log this appendix summarises. The source file was not edited at any point, so every output is a derivative and a corrected tracker can be processed by re-running the script.

What the tracker is. The CTIMP 150-day metric tracker is maintained in Research and Innovation and reported against the national targets. Its Report sheet holds one row per trial across 23 columns, of which the pharmacy block runs from the day the Research Setup and Governance meeting notifies pharmacy to the day pharmacy issues its greenlight. A second sheet, Raw data, holds 2,666 rows covering the whole portfolio including declined and closed studies, and is used only to estimate the arrival rate, since studies that never reached set-up carry no pharmacy milestones.

How the pharmacy dates get into it. By hand, from emails, local folders and the trust's quality system, with the aseptic figures arriving separately from the Pharmacy Quality Lead. None of it is a system export, so a blank records a retrieval failure rather than an event that did not happen. Two consequences. The aseptic sign-off is a tick on a whiteboard, so there is no date to retrieve. And trials before 2023 carry only HRA approval, site selected and the open-to-recruitment date, the pharmacy fields not

being collected then; since those two are the clock anchors, an old trial's deadline is computable while its duration is not.

What the tracker does not hold. Across the 104 distinct trials the cleaned file holds 28 pharmacy notification dates and 24 pharmacy green lights, no PPCTI sign-off dates at all, and 9 trials with no complexity tier. Figure B.1 sets every field against the rest. The gap is structural rather than random, but it does not fall where the obvious reading suggests. It tracks whether a record is held in a system, not which team owns the field. iQemo is pharmacy's own and is a digital system, and its dates are present for 51 of the 104 trials, better populated than 5 of the 9 fields arriving from outside pharmacy. The aseptic PPCTI sign-off, kept as a tick on a whiteboard, is empty for all 104. What cannot be retrieved is what was never written into a system, which is why chasing pharmacy would not have produced these dates and why the case studies were requested instead.

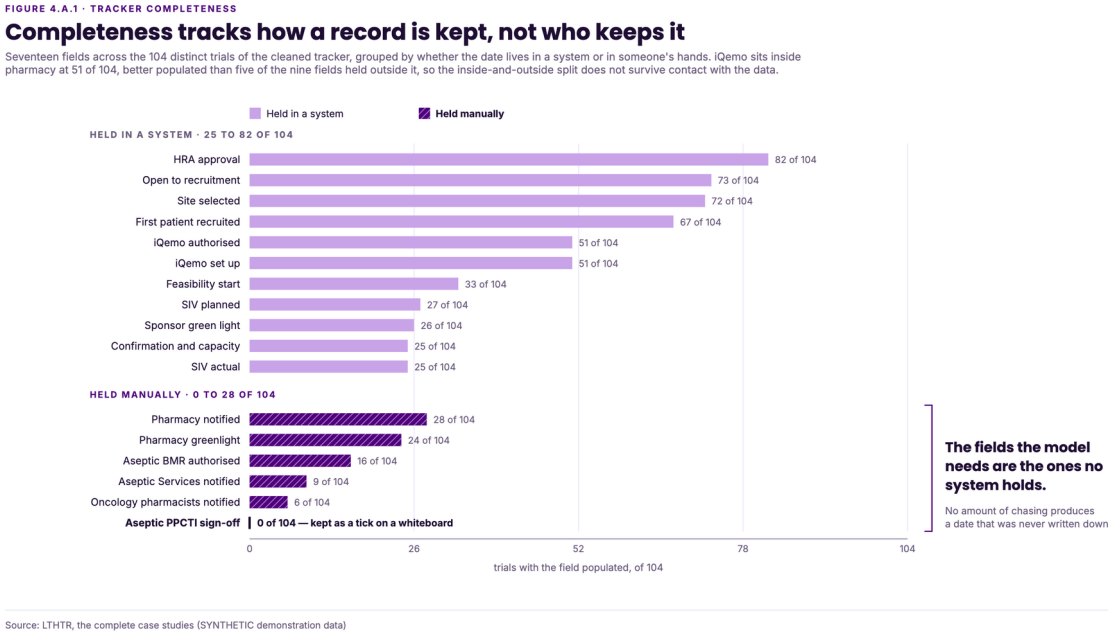


FIGURE B.1: Field completeness across the cleaned tracker, grouped by how each record is kept. Seventeen fields are shown against the 104 distinct trials of the cleaned file, split into a system-held group ranging from 25 to 82 and a manually held group ranging from 0 to 28. The aseptic PPCTI sign-off has no bar at all. A blank is a date that could not be retrieved, not a step that did not happen.

One field reported and then set aside. Confirmation and capacity to the *sponsor's* green light has a median of exactly one day across the 23 trials holding both dates. That is not a measure of governance speed. The trust's own account is that the sponsor's green light is an announcement made once everything else is complete, so the one day measures

the announcement, not the process behind it. It is a different event from the pharmacy green light of Section 4.2.2, which ends set-up; the two are months apart and only the second bounds segment C. It is recorded here so that no reader treats it as evidence that governance is fast.

Table B.1: The four faults and their treatment

Fault	**Extent**	**Treatment**
Trials stored as two half-filled rows	14 pairs	Merged on a normalised title, each field taking the single non-empty value available
"As above" in place of a date	18 cells	Replaced with the value in the row above, which is what the phrase refers to
Dates held as `dd/mm/yyyy` text	19 cells	Parsed to real dates
"Pending" in a date column	4 cells	Left blank, since it is not a date

The two positional and text faults were the more damaging of the four relative to their size. All 18 "as above" cells and 8 of the 19 text dates sat in the pharmacy columns, so nearly the whole of both notification columns was invisible to any calculation that read them as dates.

How the split rows were merged. Rows were grouped on the trial title reduced to lower case with punctuation and bracketed protocol codes removed. Within a group each field took the single non-empty value available, and the longer title was kept because it carries the protocol code. Eleven of the 14 pairs merged with no conflict at all. Most apparent conflicts elsewhere were an "as above" against a real date, or a text date against the same date stored properly, and both dissolve once the other 2 faults are corrected.

The three unresolved trials. Where the two rows of a pair disagreed on a real value, the field was left blank and logged. AMPLITUDE differs by 2 to 3 years on 4 fields, with both sets internally consistent, which suggests two different studies sharing a short title rather than one split record. TAK-079-3001 differs on 3 fields, one of which looks like a transposed day and month while the others do not. Orion-BC differs by one day on a feasibility start date and affects no model quantity. Only the second costs anything: its blanked pharmacy green light removes the trial from the sample of measurable pharmacy durations.

An internal consistency check. The tracker's observed arrival rate of 1.00 trials per month against its mean sojourn of 105.9 days implies a work-in-progress level of 3.48

Table B.2: What the clean-up changed

Quantity	Before	After
Distinct trials	118	104
Pharmacy set-up sample	24	23
Set-up median	96 days	95 days
Set-up mean	105.7 days	105.9 days
Within 60 days	2 of 24	2 of 23
Work in progress, mean	3.41	3.51
Arrival rate	0.97 per month	1.00 per month
Trials carrying a complexity tier	90.7 per cent	91.3 per cent

trials, close to the 3.51 observed after cleaning, a gap of 0.9 per cent. The agreement is descriptive only and does not validate the process assumptions used elsewhere in this study. It is the only check available on the cleaned tracker that does not depend on the fields the trust cannot supply.

What was left alone, and why. 4 items were not repaired because repair would mean invention rather than correction: the PPCTI column, empty for all 104 trials because the step is recorded as a tick on a whiteboard; 9 trials with no complexity tier, which cannot be inferred; the sparse pharmacy dates, since cleaning cannot create a record that was never made; and the three conflicts above. The start of the 60-day clock is also untouched, being a definition rather than a data fault.

B.2 Segment Statistics for the Case Studies

Section 4.2 defines the three segments and keeps only what the model consumes; this appendix holds the descriptive statistics behind that section's summary claims. Every count describes the 20 case studies and is produced by the same script as the body figures, so it regenerates whenever the pipeline is rerun.

Table B.3: Occurrence and duration of each segment across the case studies

Segment	Studies where it occurs	Median days (range)
A, preparation	5 of 19	46 (4 to 141)
B, hand-over	18 of 19	14 (−118 to 210)
C, set-up	19 of 20	79 (23 to 262)

Note. The counts are not a partition; the segments are per-study attributes. Denominators differ because one study carries no clock anchor and one carries no usable set-up start. A study with no segment A

was told after its clock had started, so no preparation window existed, which is a finding rather than a gap.

Segment A exists on 5 studies of 19. On the other 14 pharmacy was told on or after the clock-start day, so there was nothing to prepare within. Where the window does exist it is large, a median of 46 days and up to 141, with no case between 1 and 3 days. The window is therefore substantial or absent. This pattern motivates the planning-lead representation in Section 4.2.3: the preparation analyses test the effect of assigning an earlier planning lead to trials selected for commitment. The segment statistics are descriptive evidence; they do not themselves reduce the set-up duration or determine a preparation lead for every trial.

Segment B is bimodal and a single median describes it poorly. Fourteen studies have a positive hand-over, median 35 days and up to 210, consuming target days before pharmacy begins. Four are negative, at -4, -11, -33 and -118 days, meaning set-up began before the clock started. The pooled median of 14 days sits between two populations that mean opposite things, so the model carries the signed value per study rather than a central estimate, and Section 4.2.1's `handover_days` retains the tracker's 8-day figure for studies with no date at all.

Segment C has a median of 79 days, longer than the whole 60-day target before any hand-over is counted. Combined with a positive segment B on 14 of 19 studies, this is why no case study finished inside the target from a standing start, and it is the result Chapter 5 opens with.

B.3 Questionnaire-Derived Operational Inputs

Five written instruments obtained information the trust's operational systems do not hold. They went out in two waves. Parts 1 to 3 followed the meetings of 5 and 6 August 2026 and were returned between 17 and 20 August. The trial data request and the team-durations request were both dated 22 August and returned on 24 August. Table B.4 lists what each asked and where the answers enter the thesis.

The completed returns are retained in the project archive and are not reproduced here because they contain confidential operational information. LTHTR authorised the reporting of trial identifiers where required to explain the case-study findings. Only responses used as model inputs or assumptions are summarised below.

TABLE B.4: The five instruments, what each asked and where the answers are used

Instrument	**Recipient**	**Asked for**	**Used in**
Part 1	Both teams	Study identity; site selected, HRA approval, sponsor green light, date pharmacy told to start; pharmacy set-up start and green light; committee needed, submitted, cleared; electronic prescription and prescribing system, specialist input, aseptic requirement, trial phase	Clock anchors, pathway assignment, committee routing
Part 2	Research and Innovation	Prioritisation Tool scoring of studies already through pharmacy; whether the tool should recommend closing a low-scoring study; the fixed committee calendar	Basis for setting the value score aside; the committee treatment of Section 3.2
Part 3	Pharmacy Clinical Trials	Set-up duration by pathway with nothing waiting on others; lead time consumed by one sign-off; set-ups completed per month; work done before a study reaches the queue	Table B.6; work-floor audit; single-approver assumption; preparation credit
Trial data request	Both teams	For 20 tracker trials: date site confirmed from EDGE, trial phase, pharmacy set-up start, prescribing system, Stage 2 priority score, and a joint scoring of Prioritisation Tool measure 4b	Case-study parameters; the pharmacy burden rank of Appendix B.4
Team durations	Pharmacy Clinical Trials	Working days each team holds a study as an active case, by trial phase; whether the prescription build is a separate workstream; when work begins relative to notification; GMO sitting months	Per-team duration assumptions; planning-lead sensitivity; committee calendar

The trial data request and team-durations request are narrower than Parts 1 to 3. The first fills named gaps in 20 tracker trials rather than asking the service to describe itself.

The second separates one team's active-case window from the whole-service duration Part 3 reports, so the two are not alternative answers to the same question.

Table B.5: Questionnaire-derived inputs and their evidential status

Input	**Model use**	**Evidential status**
Pharmacy set-up start	Duration and reconstructed window	Retrospective estimate
Pharmacy green light	Completion endpoint	Recorded milestone
Pathway requirements	Duration basis	Study-specific reported attribute
Pharmacy burden rank	High-burden classification and the overlay	Elicited ordinal judgement
Preparation lead	Sensitivity input	Scenario assumption
Hands-on approval time	Queue sensitivity	Elicited range, not an observed time series
Delegation of approval	Queue-model caveat	Process-owner account

Table B.6: Set-up duration by pathway as stated by pharmacy, Part 3 question 1

Pathway	**Stated band**
Routine: no electronic prescription, no aseptics	Under 30 days
Electronic prescription only	Under 30 days
Aseptics only	46 to 90 days
Both electronic prescription and aseptics	46 to 90 days

Note. The two aseptic rows were answered by marking two adjacent bands; the union is recorded here.

The stated bands are used only as a service account against which observed work floors are audited, in Figure F.2; they are not substituted for the duration inputs used in the model. Two further Part 3 answers enter the thesis: the sustainable set-up rate on core hours is two studies a month, unchanged by the additional funded hours; and the pharmacy green light can be delegated to the Operational Manager once the lead is content that most processes are approved, which is why Section 4.2.3 and Appendix C treat the single-approver assumption as a conservative approximation rather than a fact.

Limits. Part 3 carries one respondent. Its answers are recalled judgements rather than measurements, and bands rather than point values. They are used as a statement of what the service believes its standard times to be, which is the only role Chapter 5 gives them.

B.4 The Pharmacy Burden Rank and the Prioritisation Tool

What the tool measures. The Prioritisation Tool, version 1.6, scores a study on eleven measures. Measure 4b, Operational Burden — Pharmacy and aseptics, is the pharmacy one. It carries five options running from −5.5 to 0, subtracting points because the tool prices set-up difficulty against a study's value. It is the finest-grained of the eleven measures and the only one that draws on pharmacy clinical trials, technical services, quality assurance, clinical and supply, and ePMA together.

Why the tool's own scores could not be used. Every study scored on the tool to date carries 4b = 0, its lowest-burden option, which reads as dispensary-led set-up of a low-complexity investigational medicinal product. Zero on every study is not a finding about the portfolio. The scoring group has no pharmacy member, so where pharmacy is not involved someone records that in free text, and where it is involved nobody in the room can judge the burden. Measure 4b is uncollected rather than flat. None of the scored studies appears in the tracker either (Section 4.1), so no tool score sits beside a set-up outcome.

What is used instead. The August 2026 return asked pharmacy to rank the case studies on pharmacy burden directly. The respondent used five distinct values, 2 to 6, on a rising scale where 6 is the most demanding. This dissertation carries that answer unchanged, as `q4b_burden_rank`. High burden means a rank of 5 or 6 throughout. The ranks are retrospective: they were given after the set-up outcomes were known, so they are a classification of what these studies demanded, not a prediction made before set-up began.

The raw pharmacy burden rank is retained as an ordinal classification. It is used for grouping, to identify high-burden studies, to interpret observed work-floor evidence, and to apply the conditional 63-day adjustment where a study is both high burden and on the Both pathway. It does not enter a Prioritisation Tool total, a value objective or an eligibility criterion, and it is never combined with the value score v_i. Keeping the raw response also keeps it auditable, because a reader who later obtains the five option values can map it, and nothing computed here would have to change.

A derived scale is used for reporting portfolio composition, and for nothing else. Comparing the burden of two committed portfolios by counting trials assumes every trial costs the service the same, which the work-floor evidence contradicts. For

that comparison alone, a derived routine-trial-equivalent planning scale maps the burden bands to relative weights of 1.00, 1.74 and 2.61, from the ratios of the fastest observed work intervals of 23, 40 and 60 days reported in Table F.3. This transformation is used in Figure 5.9 only. It is not a direct measure of staff time or hands-on workload, and no burden-resource budget is active in any reported optimisation run.

TABLE B.7: The derived routine-trial-equivalent planning scale

Pharmacy burden rank	**Derived weight**	**Basis**
2, routine	1.00	Reference: the 23-day fastest observed work interval
3 to 4, moderate	1.74	40/23, from the 40-day fastest observed interval
5 to 6, high	2.61	60/23, from the 60-day fastest observed interval

Note. The weights are derived from fastest observed work intervals in a 19-case set, and the band medians themselves are indistinguishable (Appendix F.2). They provide a planning scale for comparing the burden composition of alternative committed portfolios, and are not measured staff-time ratios: a weight of 2.61 does not assert that a high-burden study consumes 2.61 times the staff hours of a routine one. Chapter 6 records direct measurement of pathway-specific work as the prerequisite for modelling resource use in staff time rather than in commitment counts.

Appendix C

Assumptions Behind the Approval Wait

Equation (4.3) is the Pollaczek-Khinchine formula for an M/G/1 queue. The formula gives the exact long-run mean waiting time when its assumptions hold. Low utilisation reduces the size of the predicted wait. Table C.1 sets out each assumption, what it means, and how it applies to this service.

Table C.1: The M/G/1 assumptions checked against this service

Assumption	**What it requires**	**Evidence for this service**	**Status**
Poisson arrivals	Work reaches the approver independently, spread out over time rather than arriving in batches	A trial's clock starts at HRA approval or site selection, both set by the sponsor and the regulator rather than by the trust. Trials therefore arrive at their own pace. The fixed committees do not batch the signature, for the reason given below	Met, subject to the batching rationale below

continued overleaf

Assumption	**What it requires**	**Evidence for this service**	**Status**
General service time	Only the mean and the coefficient of variation are needed	Pharmacy has been asked to provide direct information on approval time. The formula itself does not require approval times to follow a normal, lognormal, or other specified distribution.	Met
Single server	One approver, with work queueing behind whatever they are already doing	The trust describes one approving person for pharmacy set-up. Pharmacy adds (August 2026 return) that the sign-off can be delegated to the operational manager when most processes are already approved	Approximation, conservative: a second signer can only shorten the wait
Work-conserving, non-preemptive	The approver is never idle with work waiting, and never abandons a part-finished approval	This is consistent with the described process. It also means that the mean wait does not depend on the order in which trials are approved.	Met
Unbounded queue	There is no fixed maximum number of approval requests that may wait; a trial is not rejected solely because another approval is pending	Trials wait for the signature. None are abandoned for want of one	Met
Independent service times	The duration of one approval does not depend on the duration of other approvals or on the number waiting	Not observable in the available records.	Assumed, recorded as a limitation
Steady state, $\rho < 1$	On average, the approver can complete work faster than new approval requests arrive.	The commitment cycle resets monthly, so the service may not fully settle into a long-run average.	Assumed, recorded as a limitation

The coefficient of variation, κ, is not identified by the available records. Equation (4.3) requires the variability of pharmacy sign-off time, but no direct observations of sign-off are available. The reported model runs therefore use $\kappa = 1.55$, the coefficient of variation calculated from 51 iQemo template-build records as the most complete pharmacy-related field in the tracker. This is a temporary substitution, not a validated estimate of approval-time variability.

The iQemo measure records the time from template set-up to authorisation. Its median is 16 days, and much of that interval is likely to be time spent waiting for the template to be processed. The required approval-time measure, by contrast, concerns the duration of the signing activity itself. The proxy therefore measures a combined waiting-and-signing interval rather than signing time alone.

The direction of the error cannot be established. The observed interval combines waiting and signing time, and adding waiting can change both the mean and the variance. Consequently, the coefficient of variation of the combined interval may be lower than, similar to, or higher than the coefficient of variation of signing alone.

This depends on the relative variability of waiting and signing time, and on whether they move together. Neither is identified in the records. Independence cannot be tested and should not be assumed.

The central problem is identification. The 16-day iQemo interval combines waiting and signing, but the available data do not separate those components. Its coefficient of variation therefore does not identify the coefficient of variation of signing time. The true signing-time κ could be lower than, close to, or higher than 1.55. The proxy consequently places little constraint on κ.

The value is retained solely to allow equation (4.3) to be evaluated while direct pharmacy sign-off data are unavailable. It is not treated as an estimate of signing-time variability, and no substantive conclusion in this thesis depends on $\kappa = 1.55$ being accurate.

The observed proxy may not vary in the same way as sign-off time. Direct sign-off observations would be needed to test whether the two are comparable. Pharmacy has therefore been asked for its quickest, usual, and longest hands-on sign-off times. These responses will define plausible bounds rather than provide an exact estimate, and the model will test gate waiting time across those bounds.

Pharmacy's three responses give a minimum-to-maximum range. However, equation (4.3) requires the coefficient of variation, κ, which depends on both the mean and the standard

deviation. A range measures the distance between the extremes, not the typical spread of observations around the mean. The range must be converted into an approximate standard deviation before κ can be estimated.

Because direct approval-time observations are unavailable, the stated operational range was converted into an approximate spread for scenario construction. The approximate standard deviation was set to one quarter of the stated range. This is a pragmatic scenario-setting rule rather than an empirically validated estimate of approval-time variability, and approval times are unlikely to be normally distributed. It is used because the available evidence consists only of three broad responses from LTHTR.

Chapter 5 currently uses $\kappa = 1.55$, because the pharmacy response was not available when the model runs were produced. Once the response is received, the model will be rerun at both ends of the plausible κ interval. If the capacity conclusion does not change, the substitution has not affected the result. If it does change, this will show that the conclusion is sensitive to variation in an approval process that the service has not yet measured.

The conversion has three main limitations. First, if approval times are right-skewed, a normal-based conversion may be misleading. Second, the number of observations behind the recalled quickest and longest times is unknown. Third, in a skewed distribution, the range may not be symmetric around the mean. A fitted lognormal or gamma distribution would be preferable, but this would require actual sign-off observations rather than three recalled bands.

A separate unresolved input is A, approver availability per cycle (capacity). It is held fixed because the trust has not supplied the required information, and this assumption is documented in Section 4.2.

IDEAS and GMO committees release studies in batches, which might be conflicted with the model's arrival assumption. However, approval occurs after a long and variable set-up period. That period disperses studies from each committee batch across later months, so much of the original batch pattern is lost before studies reach the approval gate.

This reasoning depends on approval occurring near the end of set-up. It also does not mean that the 95-day duration and the κ estimate should be treated as parameters from the same distribution: they are drawn from different samples and should remain separate.

Appendix D

Model Implementation, Verification and Run Settings

D.1 What the 73 Automated Tests Check

Section 4.4 reports 73 tests. This section says what each one checks, so the count can be verified and any single check found. Every test runs whenever the code changes, and a failing test stops the change rather than reporting the problem afterwards.

D.1.1 The selection model, tests 1 to 35

The model of Chapter 3, and the two refinements of Section 4.2.

Tests 9, 16 and 17 exercise the paused portfolio-value capability described in Section D.2.2. It is inactive in every reported run, so these three tests guard code that no reported analysis runs.

Table D.1: Tests 1 to 13: the selection model

	The test checks that
1	The demonstration instance loads with 16 trials covering all four pathways
2	The demonstration durations are short enough to show the on-time mechanism, which the measured durations are not
3	The solver finds the same plan as checking every possible plan one at a time

	The test checks that
4	The solver agrees with the simpler ranking rule wherever that rule is known to be correct
5	No plan commits more trials than the limit allows
6	The model commits fewer trials than the limit when an extra trial would cost a rescue
7	Ranking on lateness alone rescues no trial, which is why on-time is ranked first
8	Putting on-time first never improves lateness; it can only cost lateness
9	A study scoring below the value floor is closed rather than left in the queue; the floor is a paused capability, inactive in every reported run
10	An empty queue returns a plan rather than an error
11	The capacity check made before solving agrees with what the solver returns
12	A trial that cannot be planned is returned marked as blocked, not quietly dropped
13	Solving through the planner interface gives the same plan as using the model directly

Table D.2: Tests 14 to 17: the withdrawn value and trade settings

	The test checks that
14	The withdrawn setting has been removed from the model, not merely set to zero
15	A request still using it is refused rather than silently given an unchanged plan
16	Switching the value objective on never costs an on-time trial
17	Portfolio value rises with the commitment limit, which is why trading a trial away never pays

Table D.3: Tests 18 to 23: segment B, the days lost before a trial reaches pharmacy

	The test checks that
18	A trial already in pharmacy's queue does not lose the same days twice
19	A trial not yet handed over loses the days before hand-over

	The test checks that
20	A trial R&I still holds is planned on the median hand-over and marked as assumed
21	A trial with no usable dates is blocked rather than given an invented window
22	Subtracting the hand-over can cost a trial its rescue, which is the point of the correction
23	An assumed hand-over is always reported as assumed, never as a recorded date

Table D.4: Tests 24 to 35: segment A, the preparation credit

	The test checks that
24	With no credit set, every plan matches the plan produced before the credit existed
25	With both conditions met, the credit shortens the set-up
26	Complexity alone earns nothing without the sponsor green light
27	The green light alone earns nothing without complexity
28	A Phase 1 study earns the credit without needing aseptic work
29	Every input to the credit comes from the data request; none is invented
30	The planner can add a study the rule would miss
31	The planner can withhold the credit from a study the rule would give it to
32	The planner cannot waive the sponsor green light
33	The credit never shortens a study below one day
34	A negative credit is refused
35	The credit can turn a missed deadline into a rescue

D.1.2 The robustness experiment, tests 36 to 40

Tests 36 and 37 are the two hand-worked checks of Section 4.4.

Table D.5: Tests 36 to 40: the robustness experiment

	The test checks that
36	The four-trial instance of Appendix A gives the same answer worked by hand and computed by the model
37	The lateness calculation matches equation (3.5) worked by hand
38	A scenario regenerates exactly from its seed
39	The comparator commits the longest-waiting trials, as Section 4.3.1 describes
40	Checking every plan one at a time and solving directly return the same plan

D.1.3 The plan-level uncertainty run, tests 41 to 52

Tests 41 to 43 fix the run's agreement with the plan it qualifies.

Table D.6: Tests 41 to 52: the plan-level uncertainty run

	The test checks that
41	With no variation in duration, the run returns the planning model's on-time count exactly
42	With variation, the promised count is not always achieved, which is the point of the run
43	Only the set-up spread widens the drawn durations; the approval-time figure cannot
44	The reported distribution adds to one and never exceeds the number of trials committed
45	The pessimistic figure never reads higher than the average
46	The trial most likely to miss its deadline is reported first
47	A trial the plan cannot rescue is never shown as rescued
48	A larger plan carries a longer approval wait
49	A run repeats exactly from the same seed
50	A different seed gives a different result
51	An empty plan is refused
52	A run must have at least one repetition

D.1.4 Planning lead active in preparation analyses, tests 53 to 60

Added with the signed lead of Section 4.2.3. Preparation relieves the clock and must never alter the work.

Table D.7: Tests 53 to 60: the planning lead

	The test checks that
53	A planned lead extends the available window by exactly the lead
54	A zero lead leaves the window at the target
55	A negative lead, a planned late hand-over, shrinks the window by that much
56	A lead longer than the time to the clock start is clipped at the planning date, so a plan cannot bank days already passed
57	The lead never changes the tier duration
58	The lead never changes the effective duration, so capacity is untouched
59	A recorded queue-entry date always overrides a planned lead
60	A planned lead is flagged as assumed, and an observed lead is derived from the dates

D.1.5 The case-study ordering checks, tests 61 to 73

The rules that stop an impossible date ordering entering the derivation, and the segment split itself.

Table D.8: Tests 61 to 73: the case-study ordering checks

	The test checks that
61 to 63	A missing anchor, a green light before the set-up start, or a notification after the set-up start each stop the run
64 to 66	A set-up start before the clock start is **recorded, not refused**, returned in row order, and absent when the dates are ordinary
67 to 70	Segment A floors at zero while keeping its signed value; segment B carries its sign; segment C is absent rather than zero while a study is unreleased
71 to 73	The three segments reproduce the study's own clock interval, and the split is identical whichever module computes it

Note. Test 64 exists because the rule it relaxes was previously a hard failure. Rejecting a negative segment B would have discarded the four studies that met the target, which is the finding of Section 5.2.

D.2 Model Parameters, Excluded Quantities and Run Settings

This section holds the reference tables moved out of the body in the fourth draft. Nothing here is new; Sections 4.2 and 4.3 state the reasoning and this appendix carries the enumeration.

Table D.9: Model parameters and where each comes from

Parameter	Meaning	Source
p_i	Set-up duration by pathway	Duration basis selected for the analysis: case-study median regimes, demonstrated pathway floors, or pooled 62-day benchmark, as Table 4.2 specifies
ℓ_i	Preparation lead, in days	Equation (4.2). A scenario input, zero in the base case
due_i, δ_i	Due day and days remaining	Equation (3.1) from site selected and HRA approval
α_i	Days available to work	Equation (4.1), which replaces δ_i in the on-time test
a_i, w_i	Days waited and claim weight	Queue entry to planning date, then Equation (3.2)
g_n	Approval wait at n committed	Equation (4.3), tabulated before the model runs
c	Commitment limit	Set by the service, varied in Section 4.3
L	Planning cycle	30 days

D.2.1 Active planning inputs

Set-up durations. A finished case gives one observation, the days from the trial entering the pharmacy queue to pharmacy issuing its green light. Where a pathway holds fewer than 3 observations, its duration falls back to the pooled median and the fallback is reported with the figure.

Waiting time. Days waited run from queue entry, as Section 3.3 defines. The case studies provide both dates; where only notification is available, as in the tracker, it stands as a fallback proxy for queue entry.

Available window and approval wait. For each trial, the model evaluates the available pharmacy window α_i, set-up duration p_i, and approval wait g_n associated with commitment size n. The commitment limit c, planning-cycle length L, due date due_i, nominal remaining days δ_i, and lateness weight w_i are defined in Section 4.2 and used as specified in the reported analyses.

D.2.2 Recorded but excluded quantities

TABLE D.10: Quantities recorded but not used as active model parameters

Parameter	Meaning	Source
v_i	Value score	The Prioritisation Tool as recorded, supplied rather than derived
t	Eligibility threshold	Set by the service, varied in Section 4.3

Value score and eligibility threshold. The value score v_i and any associated eligibility threshold are not active model parameters. The reported formulation contains two objectives only: on-time count and weighted lateness. Prioritisation Tool scores are retained as source information but are not derived or used in optimisation, because the available score coverage does not support reliable discrimination between trials and the tool cannot be reconstructed retrospectively for studies already in set-up. Value score and priority band are distinct quantities; the latter is a coarser descriptive classification. A value-based objective would be reconsidered only if score coverage became sufficient to support a defensible ranking, not through a change to the current formulation.

Committee route. Committee requirements are recorded for contextual purposes but are not represented as optimisation constraints. The available evidence does not support a forward committee-calendar model: committee routes are incomplete, no forward calendar was available, and the observed clearances preceded pharmacy green light by 32 to 77 days. The reported model therefore does not use committee identity, meeting cadence or clearance date to select a portfolio.

Planner warning. Where the planner identifies a relevant committee for a committed trial, the tool may return a display warning stating the relevant cadence and the potential

cost of a missed meeting. The warning is advisory only: the solver does not read it and no Chapter 5 result depends on it. Committee-gate modelling remains future work, conditional on a forward calendar and evidence from cases that have passed through the relevant route.

D.2.3 Planning lead in implementation

The preparation analyses apply a planning lead ℓ_i to the available window rather than reducing the set-up duration. For a candidate plan with committed set C, each committed trial $i \in C$ is evaluated using $\alpha_i + \ell_i$; each trial outside the committed set is evaluated using α_i. This applies preparation only to trials the service has chosen to take on, which is the counterfactual examined in the preparation analyses.

The lead is subject to two constraints. First, the planned work start is floored at the planning date, so earlier preparation cannot bank time that has already passed. Second, the lead does not alter the set-up duration p_i or the commitment limit c; it changes the available window only. Section D.1.4 of Appendix D.1, tests 57 and 58, verifies that the lead alters neither set-up duration nor effective duration.

The lead is capped by the planning date, so the model cannot credit preparation time that has already elapsed before the decision is made. In the base case $\ell_i = 0$; it is varied only in the preparation analyses of Section 4.3, and granted only to the trials a plan commits, since preparation is spent on studies the service is about to take on. **Every preparation result in Chapter 5 is produced by this mechanism.**

D.2.4 Inactive duration-credit capability

Table D.11: The retained but inactive duration-credit parameter

Parameter	Meaning	Source
p_i^*	Residual duration after the duration credit	Equation (D.1). Inactive in every reported run, so $p_i^* = p_i$ throughout

The implementation retains a separate duration-credit capability, defined by Equation (D.1):

$$p_i^* = \max\bigl(1,\ p_i - k \cdot \mathbb{1}[\text{sponsor confirmed}_i \wedge \text{complex}_i]\bigr) \tag{D.1}$$

Here p_i^* is the residual duration remaining after any pre-window preparation, not a revised estimate of total set-up work. Both conditions would be required, complexity meaning aseptic work or Phase 1, and the credit k defaults to zero because early preparation is discretionary and unresourced. Its cap is zero in every reported analysis, so $p_i^* = p_i$ throughout. No Chapter 5 result uses the duration credit, and no reported run combines it with the planning lead of Section D.2.3.

The equation is displayed here rather than in the body because Section 4.2.3 states only the active preparation mechanism, the planning lead of Equation (4.2).

D.2.5 Solution method and standard run settings

For the historical replay and lever-sensitivity analyses, the reported plan is the proven-optimal solution under the stated lexicographic objective. The trade-off analysis in Section 5.6 additionally enumerates the constrained optimum at each achievable on-time count, minimising weighted lateness subject to that fixed count. These are count-constrained optima used to display the trade-off, rather than plans that are globally optimal in the lexicographic sense.

Each reported planning run solves the binary selection model defined in Chapter 3 using the active input values listed in Section D.2.1. The model selects a committed set C from the studies available at the planning date, subject to the maximum commitment limit c and the timing conditions imposed by each trial's available window, set-up duration and approval wait.

The reported formulation has two lexicographically ordered objectives:

1. maximise the number of committed trials that can complete within their available window; and
2. among plans with the same on-time count, minimise weighted lateness across the queue.

No value objective, value eligibility threshold, committee-gate constraint or burden-resource budget is active in any reported run. These quantities are either recorded for

context, returned as planner warnings, or retained as inactive implementation capabilities, as Sections D.2.2 and D.2.4 describe.

For each trial i, the model evaluates whether the active set-up duration p_i and approval wait g_n can fit inside the available window α_i. Where the preparation analysis is active, the planning lead ℓ_i is added to the available window of committed trials only, as Section D.2.3 specifies. The lead does not alter p_i, g_n, or the commitment limit c.

A trial $i \in C$ is classified as on time under a candidate committed set C of size n when its available window accommodates both the active set-up duration and the approval wait at that commitment level:

$$\alpha_i + \ell_i \;\geq\; p_i + g_n$$

with $\ell_i = 0$ in every run except the preparation analyses. The on-time count of a plan is the number of committed trials meeting this condition, and weighted lateness is summed across the whole queue, so a deferred trial contributes lateness even though it is not committed.

D.2.6 Simulation and comparator reporting rules

The two simulations answer different questions on different data and are reported separately. The plan-delivery simulation asks how reliably a selected plan delivers what it promises, drawing each committed duration from a lognormal anchored on its planned median. It runs on the real February queue and on the synthetic standing queue, because the real queue promises too little to test a promise. The robustness sweep asks whether the rule beats arrival order once the portfolio changes, and generates portfolios to do it. The first is uncertainty about one plan; the second is robustness of a rule. Neither exercise estimates the future LTHTR portfolio: each is a stated scenario analysis with a distinct purpose.

D.2.6.1 Comparator robustness sweep

The comparator robustness sweep tests the selection rule against the service's current arrival-order practice. Pharmacy has no formal rule for allocating scarce set-up capacity and generally takes trials in the order in which they have waited. The comparator

therefore commits the longest-waiting eligible trials first, up to the same commitment limit c used by the model.

Both the model and the arrival-order comparator are evaluated using the same available windows, set-up-duration regime, approval wait and commitment limit. They differ only in the committed set. Any difference in on-time completion can therefore be attributed to the selection rule rather than to a difference in the constraints applied to the two plans.

The sweep generates portfolios that vary arrival timing, pathway composition and the duration regime. It tests whether the model's advantage over arrival order survives conditions other than the single historical combination observed in the case-study replay. Generated portfolios begin with studies already in the pharmacy queue. The sweep therefore does not test the effects of preparation lead, hand-over intervention or earlier notification.

Table D.12: Robustness-sweep scenario grid

Axis	**Values**	**Rationale**
Arrival timing	Generous, typical, or little time remaining	Spans the conditions observed in the tracker, including studies already close to or beyond their due date
Aseptic-work share	0.4, 0.6, 0.8	Centres on the trust estimate that approximately 60% of open trials involve aseptic work
Prescription-only share	0.2, 0.4	Tests uncertainty in prescription-build prevalence, for which the trust has no direct portfolio estimate
Duration regime	0.8, 1.0, and 1.2 times the case-study medians; demonstrated pathway floors; pooled 62-day benchmark	Tests whether selection performance survives durations faster or slower than current performance, as well as the demonstrated-achievable planning condition

The grid contains $3 \times 3 \times 2 \times 5 = 90$ scenario cells. One hundred generated portfolios are evaluated in each cell, giving 9,000 portfolios in total under a fixed seed. The prescription-only pathway uses a pooled duration estimate where pathway-specific evidence is insufficient; findings involving that pathway should therefore be read with additional caution.

For each portfolio, the primary outcome is a strict model win:

$$N_{\text{model}} > N_{\text{arrival}},$$

where N_{model} and N_{arrival} are the respective numbers of trials placed inside the available time window. The results also report ties and the difference in weighted lateness. Results are retained at scenario-cell level rather than pooled across all portfolios, so that the timing and duration conditions associated with each outcome remain visible.

Reporting rules for ties and lateness. At a given commitment limit the model maximises the number of trials completed on time, so it cannot place fewer trials inside the window than another feasible plan; reporting that it never performs worse would describe a property of the formulation, not an empirical finding. A tie therefore has two possible meanings: the two plans select the same number of trials that finish within the window, or no trial can finish within the window at any feasible plan size, so every plan has an on-time count of zero. The second does not show that the selection rule has no value, which is why results stay at scenario-cell level.

Weighted lateness may favour arrival order in some cells, and those cells are reported rather than omitted. The objectives are ordered rather than combined into one weighted score, so the model prefers a plan that rescues one additional trial even if it produces more weighted late days. Reporting only the on-time improvement would misrepresent that trade-off.

D.2.6.2 Plan-delivery simulation

The plan-delivery simulation evaluates the reliability of one selected plan when realised set-up durations differ from the planned value. It asks whether a plan that promises a stated number of on-time trials is likely to deliver that outcome under duration uncertainty. It is run on two queues: the observed February queue, which evaluates a historical plan; and the synthetic standing queue specified in Appendix E.1, which supplies a plan with a non-zero on-time promise. The synthetic standing queue is used only for the plan-delivery reliability analysis in Section 5.3. It is not used in the observed-queue lever sensitivities of Sections 5.4 to 5.6 and it is not used in the generated-portfolio comparator robustness sweep.

For each committed trial $i \in C$ and simulation repetition r, the realised set-up duration is drawn from a lognormal distribution $D_{ir} \sim \text{Lognormal}(\mu_i, \sigma_p^2)$, where $\mu_i = \log(p_i)$ and $\sigma_p^2 = \log(1 + \text{CV}_p^2)$. This parameterisation gives the simulated duration a median equal to the active set-up duration, $\text{median}(D_{ir}) = p_i$, with mean $\mathbb{E}[D_{ir}] = p_i\sqrt{1 + \text{CV}_p^2}$.

The lognormal distribution is used because set-up durations are positive and plausibly right-skewed. Anchoring the draw on the median is important because the case-study duration regimes are represented by medians or demonstrated planning floors. Anchoring on the mean would systematically shift simulated durations away from the duration basis used to construct the plan. The duration-credit capability of Equation (D.1) is inactive in every reported analysis, its cap being zero, so $p_i^* = p_i$ and the simulation draws around the active set-up duration.

A committed trial is on time in repetition r if its drawn duration and the approval wait associated with the committed count both fit inside its available window, $\mathbb{1}_{ir} = \mathbb{1}[\alpha_i \geq D_{ir} + g_n]$. For a committed set C containing n trials, the simulated on-time count is $N_r = \sum_{i \in C} \mathbb{1}_{ir}$. Across R repetitions, the distribution of N_r estimates the probability of delivering each possible number of on-time completions. The proportion of repetitions in which trial i is on time estimates its individual on-time probability under the selected plan.

Table D.13: Settings for the plan-delivery simulation

Setting	**Value**	**Basis**
Repetitions, R	10,000	Provides a stable estimate of a probability near one half to approximately half a percentage point
Duration distribution	Lognormal	Positive, right-skewed duration draws
Draw anchor	p_i	Active set-up duration by pathway; $p_i^* = p_i$ in all reported runs
Set-up coefficient of variation, CV_p	0.467 for case-study medians; 0.575 for demonstrated floors	Measured on the same duration basis as the draws it spreads
Approval wait	g_n, fixed at committed count n	Equation (4.3)

Setting	Value	Basis
Seed	Fixed before each run	Allows all reported figures to be regenerated exactly

The simulation varies set-up duration but holds the approval wait g_n fixed at its expected value for the committed count. It therefore represents uncertainty in pharmacy set-up duration, not uncertainty in the approval-signature process. Since approval wait also varies in practice, the true distribution of plan outcomes may be wider than the distribution reported here.

Reliability thresholds. The preparation leads reported at 50, 75 and 90 per cent reliability in Section 5.5 are the corresponding quantiles of the lognormal duration distribution above, taken at the demonstrated-floor coefficient of variation, less the trial's available window. They are checked against a Monte Carlo draw from the same distribution. A reliability is therefore the proportion of repetitions in which the plan delivers the stated outcome, and the reported lead is the smallest that reaches that proportion.

D.2.6.3 Interpretation of wins, ties and lateness

At a given commitment limit, the model maximises the on-time count before minimising weighted lateness. It cannot therefore select fewer on-time trials than another feasible plan evaluated under the same conditions. A strict comparator win occurs only where the model places more trials inside the available window than arrival order.

A tie has two possible meanings. First, the two approaches may select different portfolios that produce the same on-time count. Second, no feasible portfolio may be capable of placing any trial on time under the scenario conditions. The latter does not show that the selection rule has no value; it may instead indicate that the available windows are already too short for any feasible selection to succeed. Reporting results by scenario cell preserves this distinction.

Weighted lateness may favour arrival order in some scenario cells. This does not contradict a model win on on-time completion. The objectives are lexicographic rather than combined in a weighted score: the model accepts greater weighted lateness when doing so rescues one additional trial. The reported tables therefore show strict on-time wins, ties and weighted-lateness differences together, rather than presenting only the

measure on which the model performs best. For clarity, weighted-lateness difference is reported as $\Delta W = W_{\text{model}} - W_{\text{arrival}}$. Thus $\Delta W < 0$ means that the model has lower weighted lateness than arrival order, whereas $\Delta W > 0$ means that arrival order has lower weighted lateness. This convention makes visible the cost, where present, of prioritising an additional on-time completion.

D.2.7 Lever sensitivity design

Each lever is varied on its own with the others held at base, across all twenty-five decision points, and every plan up to the commitment limit is enumerated. Each lever is scored against its own baseline setting rather than a global one. The lever sensitivity asks what would have to change for the service to place more trials inside the target window. It is the run behind Table G.2, and this section states the settings it sweeps so that table can be reproduced.

Table D.14: The six levers, their tested settings and their baselines

Lever	**Controlled by**	**Tested settings**	**Baseline**
Preparation lead, committed trials only	R&I	0, 7, 14, 21, 35, 60 days	0 days
Hand-over gap, clock to notification	R&I	Return 0, 25, 50, 75 or 100% of each trial's own observed gap	0% returned
Pharmacy queue, notification to start	Pharmacy	Return 0, 25, 50, 75 or 100% of each trial's own observed gap	0% returned
Slots opened per month	Pharmacy	1 to 6	1
Hands-on days per approval	Approver	1.0, 0.5, 2.0, 3.0 days	1.0 day
Target window	Governance	60, 55, 65, 70 days	60 days

The two delay levers are proportional, not absolute. Each trial carries its own hand-over gap and its own pharmacy-queue wait, and those differ by an order of magnitude across the case studies. A common day grid would therefore return more days to some trials than they ever lost, and would model an intervention no service could

deliver. The sweep instead returns a fraction of each trial's own observed gap, so the intervention scales with the delay each study actually experienced. The day spans quoted in Chapter 5 for these two levers are the observed range across trials, not tested values.

Each lever is scored against its own baseline, not a global one. The lateness a lever saves is measured against that lever's first setting at the same decision point. A global baseline would misattribute unopened capacity: comparing a one-slot plan against a six-slot baseline reports the slots the run did not open as lateness the lever failed to save.

Scope of each solve. One lever is varied at a time with the others held at base. Every setting is solved at all twenty-five decision points under the strict anchor rule, on the demonstrated work floors with the high-burden overlay of Section 4.2.2, and every committed set up to the commitment limit is enumerated so the optimum at each on-time count is available. The queue at each date is the trials pharmacy had been notified of and had not yet green-lit.

Outcome definitions. Four quantities are reported per setting: the number of decision points reaching at least one on-time trial; the number reaching at least two; the best on-time count achieved at any date; and the lateness saved, defined as the **largest reduction in weighted lateness at any single decision point** against that lever's own baseline. The last is a peak, not a total across the twenty-five dates, and Chapter 5 labels it accordingly.

What the sweep cannot show. The levers are varied independently, so no combination is tested and no interaction between them is estimated. The preparation mechanism and its capacity qualification are defined in Section 4.2.3; Appendix G.3 tests the latter separately.

Appendix E

Simulation Design and Robustness Results

E.1 Construction of the Synthetic Standing Queue

Section 5.3 requires a queue capable of supporting a non-zero planned on-time outcome. The observed case-study queues cannot provide such a plan without first assuming preparation. A synthetic standing queue of sixteen studies was therefore constructed for the plan-delivery reliability simulation. It is not used in the observed-queue lever sensitivities of Sections 5.4 to 5.6 or in the generated-portfolio comparator robustness analysis.

Table E.1: What each parameter of the synthetic queue is taken from

Parameter	**Value**	**Taken from**
Size	16 studies	Large enough to fill three commitment slots several times over, small enough for exhaustive enumeration
Pathway mix	7 Both, 4 Routine, 3 Prescription only, 2 Aseptic only	The case studies' own mix, and the trust's figure of roughly 60% needing aseptic work
Durations	The Appendix F.1 work floors by pathway, with Q4b 5 to 6 studies on the Both pathway at 63 days	Appendix B.2 and the burden band of Section 4.2

Parameter	Value	Taken from
Burden rank	Assigned by pathway from the observed distribution, not drawn at random	The case studies. A uniform draw on seven studies missed the top rank entirely on the first seed, which would have left the burden overlay inert
Clock anchors	Spread 15 July to 30 August 2026	A standing queue of unplanned studies is recent arrivals; older anchors would already have entered an earlier planning cycle
Notification	0 to 21 days after each study's own anchor, so no preparation window exists by default	The case studies, where 13 of 19 were told after the clock had started
Preparation	None, except three studies whose site confirmation is still ahead	Matches the observed rate of accidental preparation

The generator is seeded, so the queue regenerates identically. Its pathway mix and burden distribution are based on the case studies, but it is a scenario construct rather than a measurement or forecast of LTHTR's future portfolio. It is used only in the Section 5.3 plan-delivery reliability analysis.

E.2 The Robustness Sweep

The sweep tests one narrow question: does the selection rule's advantage survive when arrival timing, pathway mix and set-up duration are varied? It does not test preparation or the hand-over correction, and it runs on generated portfolios rather than on the trust's own queues.

90 cells, 100 portfolios each, 9,000 in total, one fixed seed per cell.

The finding is that the rule has no advantage of its own. Four of the five duration regimes return exactly zero wins in 1,800 draws each: current performance, current performance 20% faster, current performance 20% slower, and the pooled 62-day figure. The rule gains something in one regime only, the durations pharmacy has already demonstrated, and there it wins 36% of the time. An advantage that appears in one of five regimes and vanishes in the other four is a property of the queue, not of the ordering.

Table E.2: The scenario grid

Axis	**Values**	**Grounding**
Arrival timing	early, typical, late	Days remaining at the planning date, spanning the tracker's observed spread
Aseptic share	0.4, 0.6, 0.8	Centred on the trust's own 60% figure
Prescription share	0.2, 0.4	No trust figure exists, so it is varied rather than assumed
Duration regime	calibrated at 0.8, 1.0 and 1.2 times the case-study medians; the Appendix F.1 floors; the pooled 62-day figure	Current performance, demonstrated performance, and the achievable pooled quantile

Table E.3: On-time wins against the arrival-order rule, by duration regime

Regime	**Wins**	**Draws**	**Win rate**
Calibrated, 0.8×	0	1,800	0%
Calibrated	0	1,800	0%
Calibrated, 1.2×	0	1,800	0%
Floors	**640**	**1,800**	**36%**
Pooled 62 days	0	1,800	0%

The other two axes say which queues those are. The rule wins in 75% of draws when trials arrive early, 32% at the typical arrival time and never in any of the 600 draws where they arrive late, which is the same result Section 5.2 reads off the real queues: a trial that reaches pharmacy after most of its window has gone cannot be recovered by any ordering. Its wins also fall from 45% to 24% as the aseptic share rises from 0.4 to 0.8. The trust's own mix of roughly 60% aseptic sits between them, at 38%, and the direction of travel is towards the heavier end.

Three things follow. First, selection is the third lever rather than the first. A service that adopts this rule without changing durations gains nothing, and the four zero rows are what that looks like measured; Section 5.4 reaches the same ordering from the real queues by finding preparation and the two delays ahead of any scheduling change. Second, the rule is a rationing device rather than an improvement. Section 5.1 finds its advantage on the real queues at one and two set-ups a month and gone from three, so it earns its keep exactly where capacity is tightest and stops mattering once capacity is not, which is the opposite of how a scheduling tool is usually sold. Third, each win is bought. The 640 wins cost 24.3 weighted days of lateness elsewhere apiece, so the rule

trades delay across the queue for a threshold rather than removing delay, which is the Section 5.6 trade-off appearing again on generated data.

Three limits. The model never loses, which is by construction rather than a finding: it attains the maximum achievable on-time count, so a loss is impossible. The 36% is therefore 36% better and 64% identical, not a rule that is worse the rest of the time. The generator draws studies already queued, so the sweep cannot exercise preparation and understates it throughout. And the comparator, committing the longest-waiting study first, is more mechanical than real practice, which already accelerates studies judged important; any advantage shown here is therefore an upper bound.

E.3 The Plan-Delivery Simulation on the Real Queue

This section reports the plan-delivery simulation in full: what the real February 2025 queue delivers under each duration basis, and the one deterministic margin the simulation resolves. It assesses uncertainty around selected plans; it does not test whether the selection rule outperforms arrival order, which is the question Appendix E.2 answers across generated portfolios.

The two duration bases carry different spreads as well as different medians, a coefficient of variation of 0.467 at current performance and 0.575 at the demonstrated floors, so the two rows of Table E.4 differ on both.

Table E.4: What the February 2025 queue delivers when durations vary

Duration basis	**Plan promises**	**0 on time**	**1**	**2**	**3**
Current performance	0	70.9%	29.0%	0.1%	0.0%
Demonstrated floors	0	33.0%	49.3%	17.7%	0.0%

Note. Approval wait times are held at their expected values, so every row understates total uncertainty.

Both rows promise nothing, and both therefore keep their promise in every run: a plan that commits to no on-time completion cannot fall short of one, and so cannot test whether a promise survives. What the table does show is how far the outcome moves on duration alone. At current performance one trial finishes on time in 29% of runs; at the demonstrated floors that rises to 49%, and two trials finish on time in a further 18%. The model commits the same three trials in both rows, so this is the spread around one plan rather than a different decision. None of it can be planned for: an on-time completion that the plan does not promise is luck, and the service cannot schedule against it.

The same analysis resolves the uncertainty noted in Section 5.2. BNT327-06 carries burden rank 6, so the plan gives it the 63-day high-burden floor against a 62-day window and records it as one day late. In simulation it finishes on time in 48% of runs, which is the case behind Section 5.3's closing statement that a margin smaller than the spread is a tie-break rather than a finding.

Appendix F

Empirical Duration Evidence, Work Floors and Statistical Checks

F.1 The Measured Work Floors and What Consumes the Window

This section distinguishes three related quantities. First, the burden-band minima of 23, 40 and 60 days are descriptive evidence about the slack observed within each burden group. Second, observed pathway-level work intervals are calibration evidence. Third, the pathway floors in Table 4.2 are the active duration inputs used in the reported planning analyses. The 63-day value is a conditional overlay that applies only where a trial is both high burden and on the Both pathway. These quantities are related but are not interchangeable.

Each floor is anchored on the two fastest work intervals its pathway has recorded, taking the second where a single minimum would rest on one noisy case, except prescription-only, which is set at pharmacy's stated band because its fastest case confirms it. The durations the model plans with are medians and floors of the case studies' own recorded work, not estimates. This section derives them and audits them against the bands pharmacy stated in the questionnaire. The four trials that met the target are a finding rather than a parameter and are reported in Section 5.2.

No case study met the 60-day target from a standing start. The four that achieved it did so only by starting early. Table F.1 reports the three measures defined in Section 4.2.

Table F.1: Set-up performance of the case studies

Measure	**n**	**Median (days)**	**Range**	**Within 60 days**
Clock interval (clock start to green light)	19	126	34 to 345	4
Pharmacy-held interval (notified to green light)	20	102	61 to 283	0
Work interval (set-up start to green light)	19	79	23 to 262	5

The work itself is not the main cause of delay. Figure F.1 shows that five of the 19 recorded work intervals fit within 60 days.

Delays occur before and around the work, in activities not captured by the tracker. Table F.2 lists these factors from the pharmacy team's meeting records. As none is dated in any system, recorded duration reflects both congestion and work.

The chapter therefore separates the two: the median shows current performance, while the fastest cases show what the service can achieve.

Table F.2: Recorded reasons the window is consumed outside the work (pharmacy team meeting record, 30 June 2026)

Mechanism	**Recorded example**	**Dated in any system**
Sponsor-documentation dependency blocks sign-off	An identity card waiting on vial photographs	No
Clarification loops on contradictory, jargon-heavy sponsor documents	"all these emails have to happen to check our understanding"	No
Contradictions inside the pharmacy manual itself	Thaw time stated differently in two places	No

The floors also test the durations stated by the trust. Figure F.2 compares the two fastest recorded work intervals for each pathway with the duration band pharmacy gave in the August 2026 questionnaire (Appendix B.3).

Three of the four pathways match their stated duration band. Routine does not. Pharmacy estimated that a study with no aseptic work or prescription build would take under 30 days, despite having roughly 52 days within the 60-day target. However, the

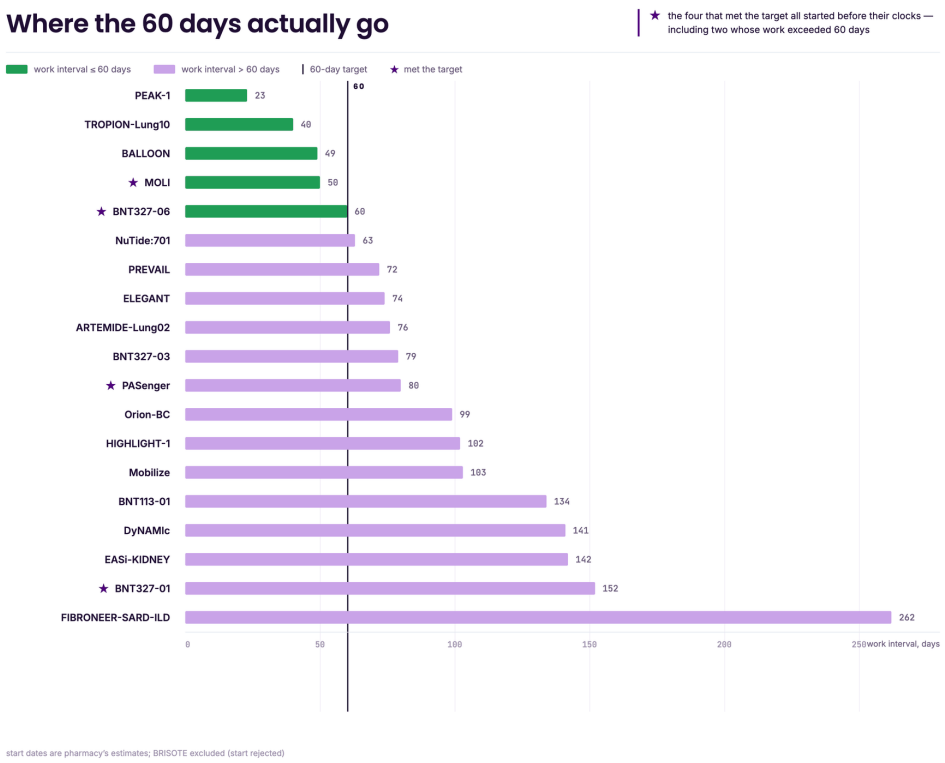


FIGURE F.1: Recorded work intervals against the target. Nineteen sorted work-interval bars are shown against a 60-day line; five fit inside it, and the four trials that met the target are starred.

two fastest routine cases, BALLOON and MOLI, took 49 and 50 days. As these were independent trials with almost identical durations, the gap is unlikely to reflect an isolated record.

Three explanations are possible, but this dataset cannot distinguish between them.

First, the recorded start date may be earlier than the actual start of work. Every set-up start date is an estimate based on recall (Section 4.2), so an early estimate would lengthen the recorded interval.

Second, the interval may include waiting time rather than work. A routine study may wait behind aseptic studies already assigned to the same staff, and the dataset does not distinguish between days worked and days open.

Third, the stated band may refer to hands-on effort rather than elapsed time. The questionnaire did not ask respondents to separate these measures.

The first two would mean routine work is faster than 49 days; the third would mean the band and the measurement are not comparable quantities. Separating them needs a further investigation for the trust.

The confirmed floors do not increase with pathway complexity. On the observed two-fastest-case measure, the combined aseptic-and-prescription pathway has a value of 40 days, below the Routine pathway. This is descriptive calibration evidence rather than the active planning input. The model uses a 60-day Both-pathway planning floor, selected to avoid relying on a single unusually fast case. The floor rises instead with the burden a study carries, on the ranking defined in Appendix B.4. Table F.3 groups the 19 recorded work intervals by burden band.

Table F.3: Recorded work intervals by pharmacy burden band

Burden band	**n**	**Fastest observed work**	**First quartile**
2, routine	6	23 days	42 days
3 to 4, moderate	7	40 days	72 days
5 to 6, high	6	60 days	62 days

Burden separates the floors, not the distributions. The fastest case increases with each burden band, from 23 to 40 to 60 days, while the quartiles show no clear order and the medians are indistinguishable (Appendix F.2).

The observed high-burden minimum reaches the 60-day target, indicating little observed slack from a standing start. This supports the planning concern but does not establish a universal duration rule for every high-burden trial. Its fastest recorded set-up already uses the full 60-day window, so it cannot meet the target from a standing start.

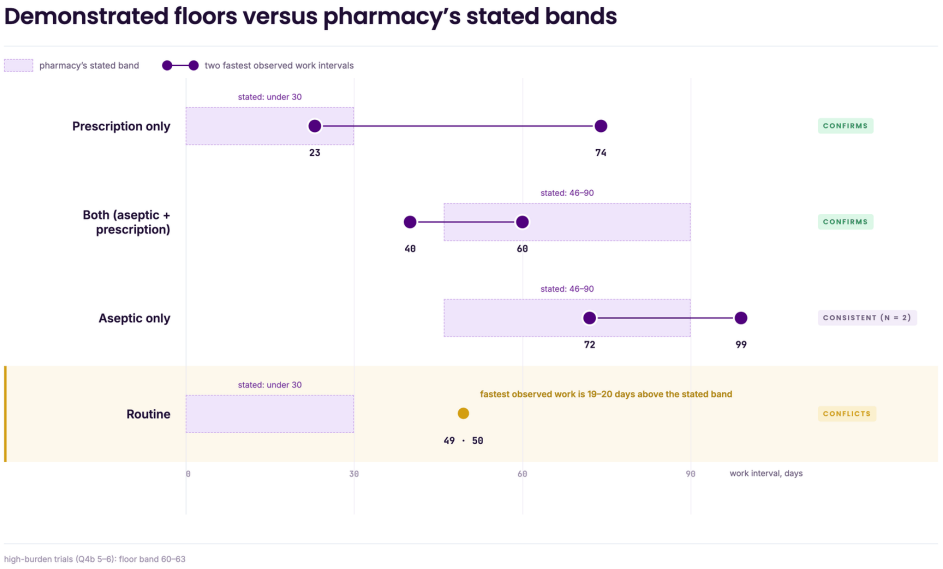


FIGURE F.2: Demonstrated floors against pharmacy's stated bands. A dumbbell chart of the two fastest work intervals per pathway is set against each stated band; three pathways agree and Routine conflicts.

F.2 Statistical Tests Behind the Work Floors

Every test below is computed on the case studies by the script `analyse_preparation_and_burden.py`. Non-parametric tests are used throughout because the samples are small and the durations are right-skewed.

TABLE F.4: Does preparation change the work, or only the clock?

Comparison	Groups	Result	Reading
Work interval, early against late starters	4 against 14	median 70 against 89 days, Mann-Whitney $p = 0.721$	No detectable difference in recorded work interval was observed between early and late starters
Met the target, early against late starters	4 against 14	4 of 4 against 0 of 14, Fisher $p = 0.0003$	Early starters were more likely to meet the target, indicating that earlier work changes how much of the process falls inside the target window

TABLE F.5: Does any recorded characteristic predict an early start?

Characteristic	Early	Late	Fisher p
Aseptic	3 of 4	8 of 14	1.000
Oncology	3 of 4	11 of 14	1.000
Phase 2 rather than 3	2 of 4	4 of 14	0.569
Burden rank 4 or above	2 of 4	7 of 14	1.000
Chemotherapy ePMA	3 of 4	10 of 14	1.000

Table F.6: Does pharmacy burden predict duration?

Test	**Result**
Burden rank against set-up total, all studies	Spearman $\rho = -0.065$, $p = 0.785$
Excluding the single Tukey outlier	$\rho = +0.038$, $p = 0.878$
Excluding the two longest	$\rho = -0.108$, $p = 0.669$
Excluding the three longest	$\rho = -0.001$, $p = 0.996$
Excluding two longest and two shortest	$\rho = -0.222$, $p = 0.408$
Burden 4 to 6 against 2 to 3	median 95 against 107 days, Mann-Whitney $p = 0.447$
Within aseptic studies only	$\rho = +0.297$, $p = 0.374$
Work interval, burden 2 to 4 against 5 to 6	median 80 against 78 days, Mann-Whitney $p = 0.831$
Work interval across the three bands of Table F.3	Kruskal-Wallis, 2 df, $H = 1.171$, $p = 0.557$

no more: continuing until one crossed significance would be searching for a result rather than testing for one.

Table F.7: Do the pathways separate on duration?

Quantity	**Test**	**Result**
Work interval by pathway	Kruskal-Wallis, 3 df	$H = 0.118$, $p = 0.990$
Set-up total by pathway	Kruskal-Wallis, 3 df	$H = 1.978$, $p = 0.577$

No statistically detectable pathway difference was observed in this small sample. The pooled median is therefore retained for the current-performance duration regime, while pathway-specific demonstrated floors are reported separately as an achievable planning condition.

On what these tests can and cannot show. With 20 studies and 4 in the early group, they can rule out a large effect and not a modest one. The null results are reported as absence of evidence at this sample size, not as evidence of absence.

Appendix G

Preparation, Delay and Capacity Sensitivity

G.1 Targeted Against Untargeted Preparation

Section 5.4 reports that preparation aimed only at high-burden studies reaches almost as many dates for about half the earlier notification. This is the sweep behind that sentence.

The regime matches Table G.2: the strict anchor rule, the demonstrated floors, the high-burden overlay at 63 days, six slots, and the same twenty-five decision points. Targeted preparation grants the lead only to committed trials in burden band 5 or 6. Spread preparation grants it across the queue.

Table G.1: What each lead reaches, targeted against spread

	Targeted				**Spread**			
Lead	≥ 1	≥ 2	**Best**	**R&I days**	≥ 1	≥ 2	**Best**	**R&I days**
0 days	9 of 25	0	1	0	9 of 25	0	1	0
7 days	10 of 25	0	1	147	11 of 25	0	1	469
14 days	11 of 25	0	1	294	12 of 25	2	2	980
21 days	12 of 25	0	1	441	14 of 25	4	2	1,407
35 days	14 of 25	1	2	735	18 of 25	7	3	2,450
60 days	16 of 25	2	2	1,260	20 of 25	12	3	4,020

Read the table across equal counts rather than equal leads. Twelve dates cost 441 trial-days of earlier notification when the lead is targeted and 980 when it is spread; fourteen

dates cost 735 against 1,407. Roughly half the R&I effort reaches the same number of dates.

The ceiling is what targeting gives up. It never reaches three trials on time, and reaches two at only two dates, against three and twelve when the lead is spread. Not every rescuable trial is a heavy one, so a lead confined to bands 5 and 6 cannot reach the trials that the spread lead rescues.

Every figure comes from `analyse_burden_and_preparation`. Its untargeted arm reproduces the preparation row of Table G.2 exactly, 20 of 25 dates at a 60-day lead, which is what makes the two arms comparable.

G.2 The Evidence Behind the Lever Table

Section 5.4 compares the six settings as a figure and states each finding in a sentence. This section holds Table G.2, which gives the tested range and the values behind that figure, and the four figures behind those sentences: the baseline every row is read against, the four settings drawn as surfaces, and the two settings whose surfaces explain why they leave the on-time count untouched.

Note. R&I controls the preparation lead and the hand-over gap, pharmacy the queue and the slots, the approver the hands-on days and governance the target window; every lever is measured except hands-on days, which is assumed, and the target window, which is policy. For the two delay levers the percentages apply to each trial's own observed delay, so the day spans are observed across trials rather than common tested values. The final column is the largest single-date reduction in weighted lateness against that lever's baseline, not a total. Appendix D.2.7 gives the full design.

Figure G.1 combines all queues, with lateness normalised by the number of waiting trials so that months with four and seven trials can be compared on the same scale. It is the baseline for Section 5.4's statement that nine of the twenty-five dates already admit one on-time trial with no further change. Those nine months are also the least late, none exceeding 85 days per trial, while months with no rescue reach up to 350.

Figure G.2 shows four of the six settings on the same axes as the preparation surface, one panel per setting. The two delays are plotted as the delay itself, so each surface falls as the delay increases.

Table G.2: What each setting changes, across twenty-five decision points

Setting	**Intervention tested / observed span**	**Months ≥1 on time (of 25)**	**Best count**	**Largest single-date lateness reduction**
Preparation lead, committed trials	0, 7, 14, 21, 35, 60 days	20 of 25	3	394
Pharmacy queue, notification to start	Return 0, 25, 50, 75 or 100% of each trial's observed wait, which spans 0 to 23 days	14 of 25	2	76
Hand-over gap, clock to notification	Return 0, 25, 50, 75 or 100% of each trial's observed gap, which spans 0 to 112 days	11 of 25	2	498
Slots opened per month	1, 2, 3, 4, 5, 6	9 of 25	1	154
Hands-on days per approval	0.5, 1.0, 2.0, 3.0 days	9 of 25	1	20
Target window	55, 60, 65, 70 days	11 of 25	2	77

Removing the entire hand-over gap increases the on-time count from nine months to eleven. Removing the pharmacy queue increases it to fourteen months, over a span one-fifth as long. In contrast, slots and hands-on approval days remain flat at nine across their entire ranges. This is the evidence for Section 5.4's statement that the two delays move more months onto the target without raising the ceiling.

The two columns of Table G.2 rank the interventions differently because on-time completion is a threshold measure, whereas lateness accumulates. Reducing the R&I hand-over gap produces the largest single-date reduction in weighted lateness, 498 days at one decision point, but increases the on-time count at only two additional decision points,

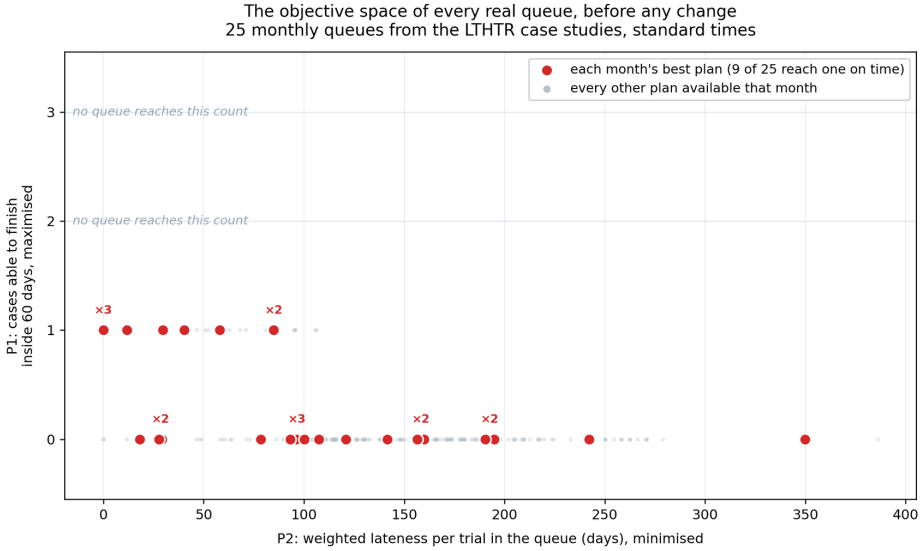


FIGURE G.1: The objective space of every real queue before any change. Every plan of all twenty-five monthly queues is plotted as weighted lateness per trial against on-time count; each month's best plan is marked in red, nine of them reach one on time, and the counts of two and three are marked as unreached.

because it mainly affects trials already beyond rescue. The pharmacy queue never exceeds 23 days and its best single-date reduction is around one-seventh as large, yet it improves the on-time count at five additional decision points, because those days affect trials close to the 60-day threshold. The target window, the only setting the trust does not control, reaches one on-time trial in 11 of the 25 months across the 55 to 70 day range tested, the same count as removing the entire hand-over gap.

Read against the preparation surface of Figure 5.5, a 10-day lead enables one on-time trial in 11 months, a 30-day lead in 17 months, and a 60-day lead in 20 of the 25 months. Two on-time trials are harder: no month reaches two below 10 days of lead, seven reach it at 30 days and twelve at 60. Three on-time trials occur in one month at 30 days and two at 60.

Figure G.3 varies the hands-on approval assumption of Section 5.4 alongside the number of set-ups committed at once, with the expected wait at the signature stage as height. The 60-day target is included as a reference plane.

Below one hands-on day, the expected wait is under one day when three trials are committed at once, and about three days at six trials. At 1.5 days, the wait reaches 48.5 days with six concurrent commitments. At two days, the 60-day target is exceeded

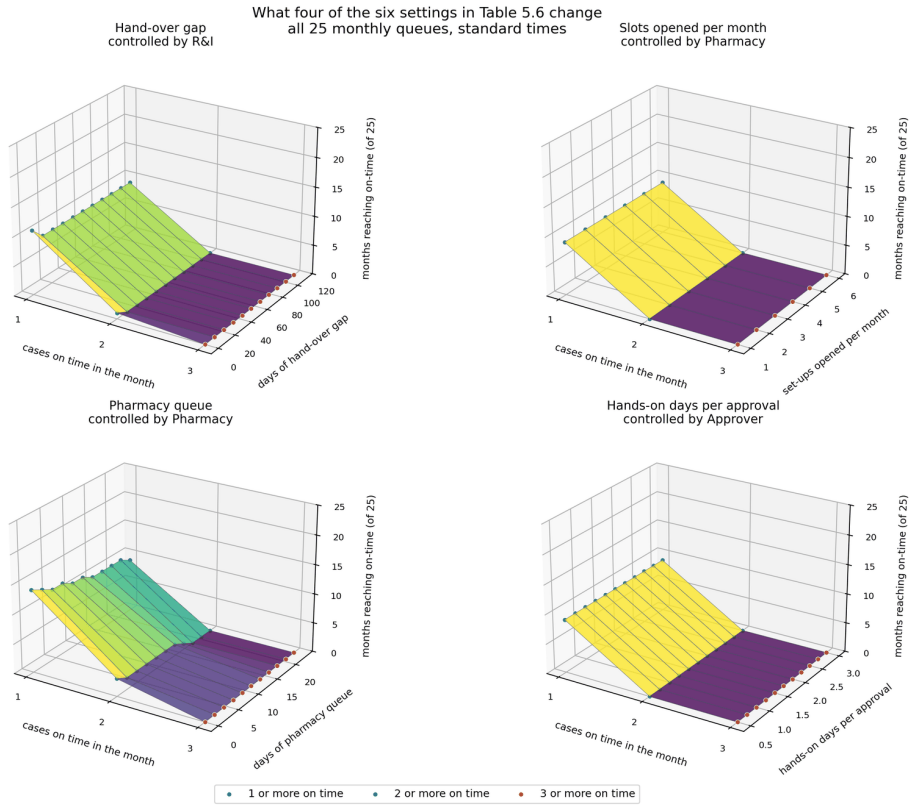


FIGURE G.2: Four settings of the lever table, on the axes of the preparation surface. Four 3-D surfaces are shown, one per setting, each over cases on time and that setting's own value with the number of months out of twenty-five as height; the hand-over gap falls from eleven to nine as the gap grows, the pharmacy queue from fourteen to nine, and slots and hands-on days stay flat at nine.

once five trials are committed at once. At three days, the expected wait reaches 97 days even with only three concurrent commitments.

When hands-on approval time is below 1.5 days, all six set-ups are worth opening. At two days, about four are worthwhile, and at three days, only two. This is the range behind Section 5.4's statement that performance is insensitive below the assumed one hands-on day and sharply sensitive above it.

Figure G.4 varies the set-ups committed against the preparation lead across the same twenty-five queues, with the burden the best plan commits as height. The surface is flat along the preparation axis: at every capacity the burden committed is the same at a 45-day lead as at none, and at 60 days it falls, from 5.2 units to 4.6 at five and six set-ups, because a long lead makes the lighter studies rescuable and the model stops

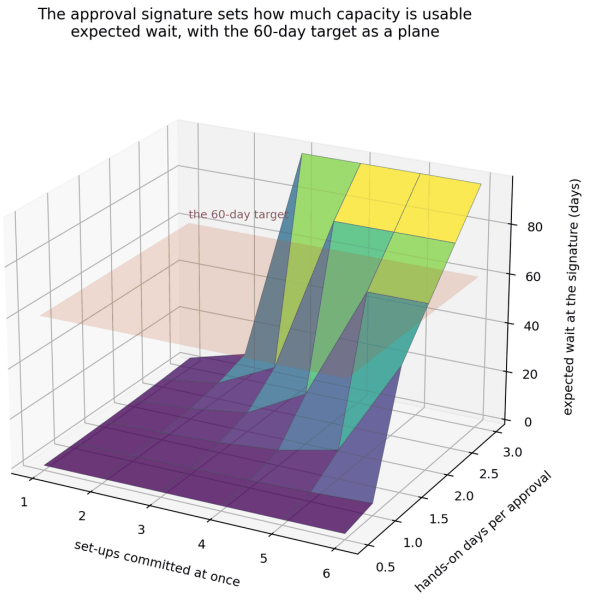


FIGURE G.3: The approval signature sets how much capacity is usable. A 3-D surface is drawn over set-ups committed and hands-on days per approval, with the expected wait at the signature as height and a translucent plane at 60 days; the surface is flat until 1.5 hands-on days and then rises through the plane.

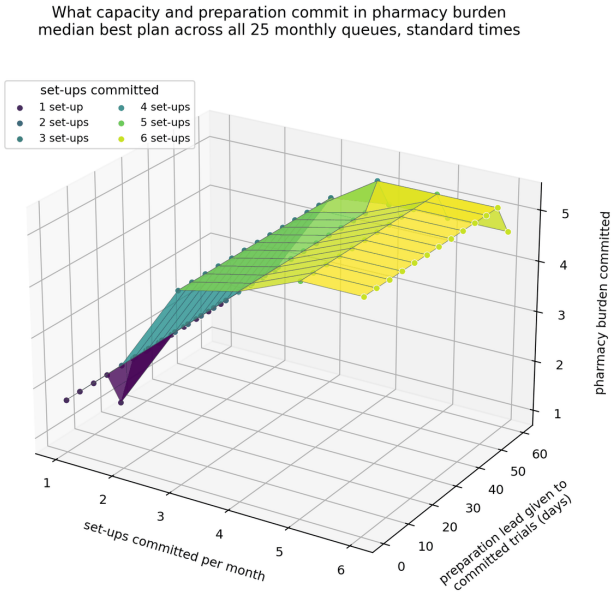


FIGURE G.4: What capacity and preparation commit in pharmacy burden. A 3-D surface is drawn over set-ups committed per month and preparation lead, with the median pharmacy burden the best plan commits as height; the surface rises with set-ups from 1.7 to 5.2 units and is flat along the preparation axis, dipping to 4.6 only at a 60-day lead.

reaching for the heavy ones. This is the evidence for Section 5.5's statement that burden is set by how many set-ups the service opens, not by how early it is told.

The frontier values behind the trade-off section. On the largest queue, 1 May 2025, a 21-day preparation lead produces three frontier plans: zero trials on time with 511 weighted late days, one trial on time with 523 days, and two trials on time with 684 days. The first on-time completion therefore costs 12 weighted late days, and the second a further 161. With a 35-day lead the plan that brings two trials on time also has the lowest weighted lateness, so those two are achieved with no lateness trade-off and only the third costs anything, 31 weighted days. On the 1 February 2025 queue the two objectives do not conflict at any preparation lead.

What capacity buys, in days and in burden. Figure 5.9 takes the median across the twenty-five queues with lateness per trial so that months of different sizes are comparable. Lateness falls with every slot opened at each on-time count, from 61 days per trial to 33 at one trial on time. Expressed in pharmacy burden rather than days, the same plans rise with both capacity and the on-time count, from 1.7 routine-trial equivalents for a single rescue at one set-up to 8.3 for three rescues at six, on the derived planning scale of Appendix B.4.

G.3 Preparation Charged Against Capacity

The preparation lever of Section 5.4 adds days to the window of every trial a plan commits and never charges the relocated work against pharmacy capacity. This section sweeps the preparation lead against the number of concurrent set-ups, so that the omission becomes a bound rather than an unknown: if preparation consumes capacity, so that fewer set-ups run at once, does it still reach three trials on time?

Note. Twenty-five decision points, April 2024 to April 2026, strict anchor rule, standard times, high-burden overlay, no synthetic data. The six-slot column reproduces Section 5.4 exactly, at 9, 11, 17 and 20 dates reaching at least one.

Three readings. The count of dates reaching at least one on-time trial does not move with capacity at any lead, which is the same finding Section 5.4 reports from the other direction. The three-trial result is the one that depends on capacity, and it survives to three concurrent set-ups and fails at two. And against 3.51 set-ups observed in progress (Appendix B.1), the margin before that failure is roughly half a set-up.

Table G.3: Decision points reaching each on-time count, by preparation lead and concurrent set-ups

Lead	**Count**	**6**	**5**	**4**	**3**	**2**	**1**
0 d	≥ 1	9	9	9	9	9	9
10 d	≥ 1	11	11	11	11	11	11
30 d	≥ 1	17	17	17	17	17	17
60 d	≥ 1	20	20	20	20	20	20
30 d	≥ 2	7	7	7	7	7	0
60 d	≥ 2	12	12	12	12	12	0
30 d	3	1	1	1	**1**	**0**	0
60 d	3	2	2	2	**2**	**0**	0

Appendix H

When the Model Would Start Each Trial

Section 5.2 reports that three of the four trials the model brings in on time are among the six it starts earliest. This appendix sets its start date against the service's own for every trial, and states the caveat that reading requires.

November 2024 shows what waiting costs. The service had TROPION-Lung10 and EASi-KIDNEY under way, neither able to finish within its window, and did not start MOLI, which could, until 5 December.

Overall the model starts only 6 of the 18 comparable trials sooner and 12 later. This is because it commits only on the first day of each month, so a trial arriving mid-month waits for the next planning cycle. Figure H.1 therefore compares a monthly planning cycle with continuous decision-making rather than measuring which approach is faster, and the twelve later starts are a property of the planning cycle rather than a result about the rule.

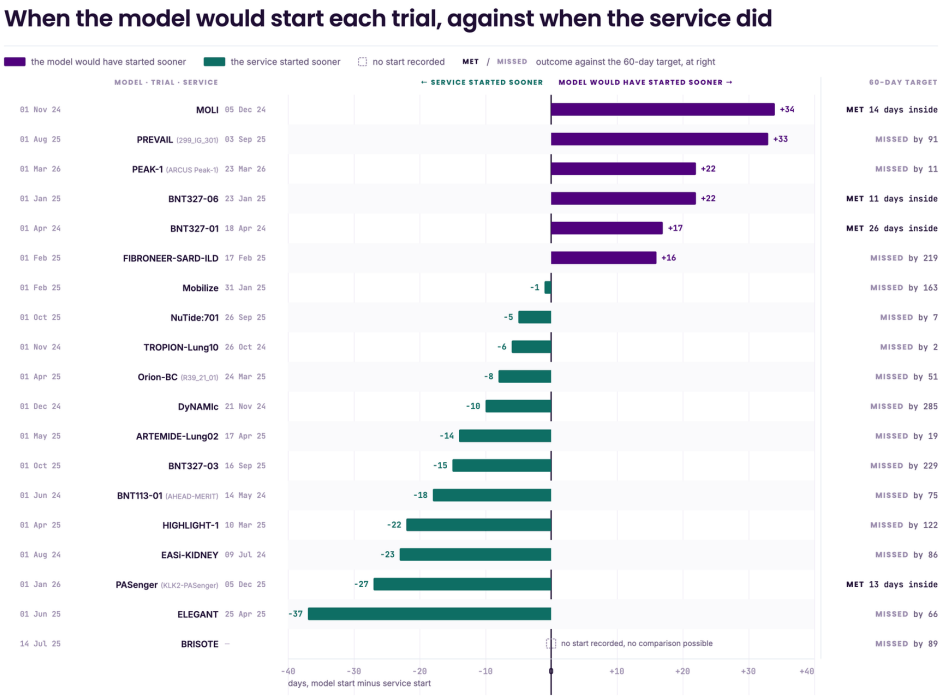


FIGURE H.1: When the model would start each trial, against when the service did. A diverging bar chart of nineteen trials is shown, each bar the difference in days between when the model would commit the trial and when the service actually started it; six run right for the model starting sooner and twelve left for the service, with three of the four on-time trials among the six.

Appendix I

What the Model Would Look Like in Use

The preceding chapters report the selection model and the evidence it produces. This appendix sets out a third contribution: an interface specification and visual mock-up showing how those inputs, recommendations and limitations could be presented to Research and Innovation and Pharmacy Clinical Trials users in a monthly planning meeting.

It is a design artefact, not a system. No solver runs behind it, it holds no trust data, and it has not been used to plan a real month. Every number on the screens is fixed example data. The appendix therefore documents not a working tool, but the design choices that determine what an eventual interface would show, make hard to miss and refuse to claim.

I.1 Keeping three kinds of statement apart

The same screens can produce three different kinds of statement, and confusing them is the main risk in putting a model in front of a service. Evidence replay reports what happened on the historical caseload. Explore scenario shows how the picture moves when a setting is changed. Live monthly planner prepares a recommendation for a month that has not happened yet.

Figure I.1 shows how the interface separates them. The mode is chosen before anything is loaded, and each card states both what its output may be used as and what it must

not be read as. The chosen mode then appears on every screen after it, as a banner and as a coloured edge, so that a single screenshot taken out of context still says which kind of statement it carries. Explore scenario carries a hatched edge, because that is the mode whose figures are easiest to mistake for findings.

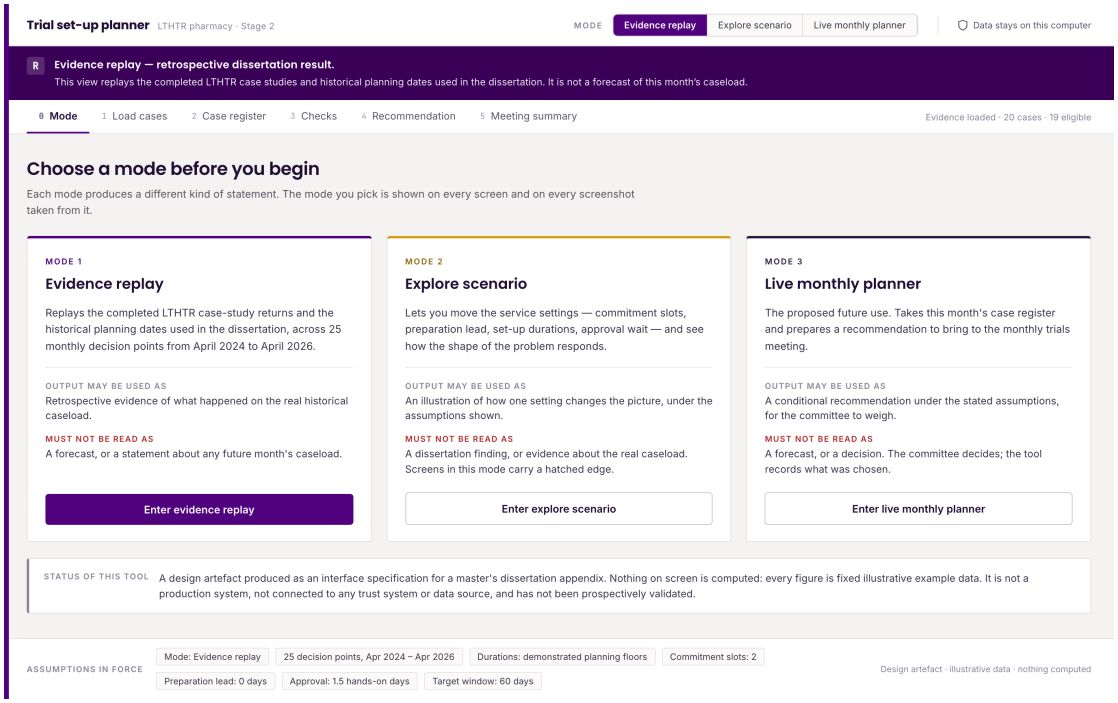


FIGURE I.1: Choosing the mode before anything is loaded. Three cards set out evidence replay, explore scenario and live monthly planner; each names what its output may be used as and what it must not be read as, and a footer states that the tool is a design artefact in which nothing is computed.

I.2 Loading a caseload

Three routes open three different kinds of data, and Figure I.2 keeps them visibly apart rather than behind one generic button. The first opens a fixed evidence-replay view based on the twenty completed case-study returns used in this dissertation. The second takes an uploaded register for a month the service is actually planning. The third starts empty, for typing cases in as pharmacy is notified.

The status panel beside them reports what arrived and what is missing. It also states the point that Appendix B.2 makes at length, which is that the denominators differ: the replay uses the nineteen cases holding both clock anchors, the work-interval evidence uses a different nineteen, and the pharmacy-held interval uses all twenty. An interface

that printed a bare 19 without saying which nineteen would be repeating an error the data made easy.

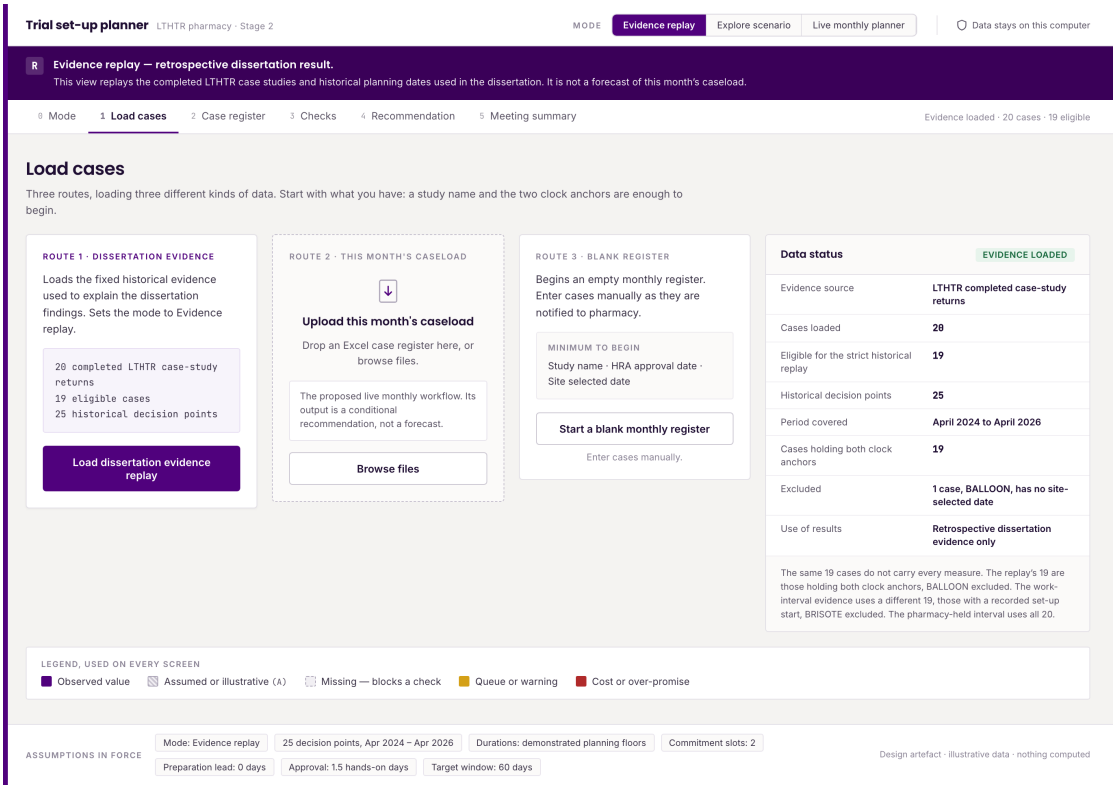


FIGURE I.2: Three routes into the tool, and what each one opens. The dissertation evidence replay, an uploaded monthly caseload and a blank register are offered separately, and the status panel names the source, the counts, the excluded case and the use the results may be put to.

I.3 The case register

The register in Figure I.3 is one row per trial. Two things on it carry most of the weight.

The first is the clock start. Both anchors are shown, HRA approval and site selected, and the later of the two is marked and bolded, because that is the date the sixty days run from. Showing one anchor, or showing both without saying which binds, is how a register can be accurate and still be read wrongly.

The second is the column for time lost before pharmacy. It is given as its own column rather than left to be worked out, because it is the quantity that decides most cases. Four rows show a negative value, meaning pharmacy had the trial before the clock started, and those four are the four that met the target.

Every assumed or illustrative value on the register is marked, and a case with a field missing is marked as blocked rather than dropped. Only five real study names appear, which are the five this dissertation already publishes; every other row is a numbered case, and every date on the screen is illustrative.

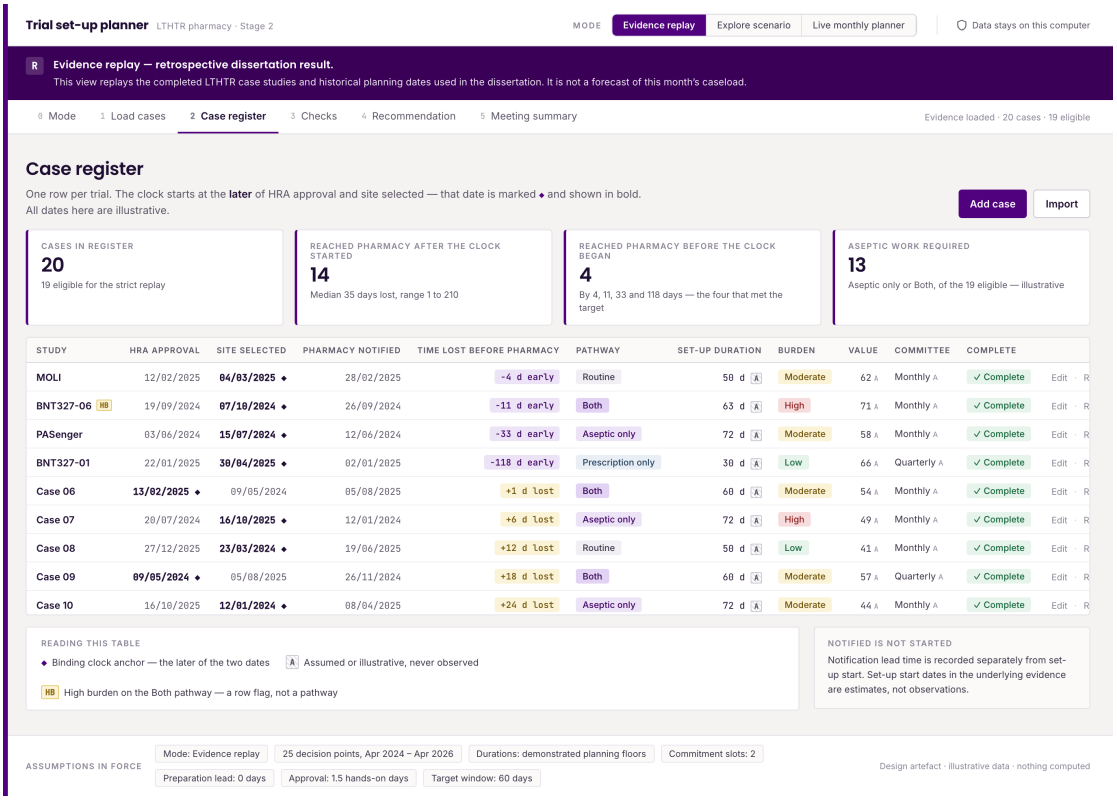


FIGURE I.3: The case register, with the binding clock anchor marked. Each row carries both anchors with the later one bolded, the pharmacy notified date, time lost before pharmacy, pathway, set-up duration with its source, burden, value score and a completeness chip.

I.4 The three checks

Figure I.4 is the screen the argument of this dissertation rests on. It reads in sequence: what the clock leaves, what the study information is for, and whether the service assumptions can fit inside what is left.

The middle card exists to prevent a misreading. Value score and committee route are labelled advisory, and burden is labelled a risk display, because none of them selects cases. Burden in particular is a slack classification built on the observed floor, and the

trust cannot decline trials in any case, so the answer to burden is preparation rather than refusal.

The third card is where the finding lands. Among the fourteen case studies that reached pharmacy after the clock started, the median hand-over was thirty-five days. This leaves twenty-five days of the sixty-day target in a representative late-arrival case, before any pharmacy work begins. Every demonstrated planning floor except Prescription only exceeds that, and every current-performance median exceeds the whole sixty-day target.

The screen states this as a warning with real visual weight rather than as a footnote, because it explains why opening more commitment slots does not raise the observed on-time ceiling: usable time remaining when a trial reaches pharmacy, rather than nominal capacity, binds on-time completion. Capacity still buys something, which Section 5.6 measures as lower lateness across the rest of the queue, but it does not buy more trials inside the window.

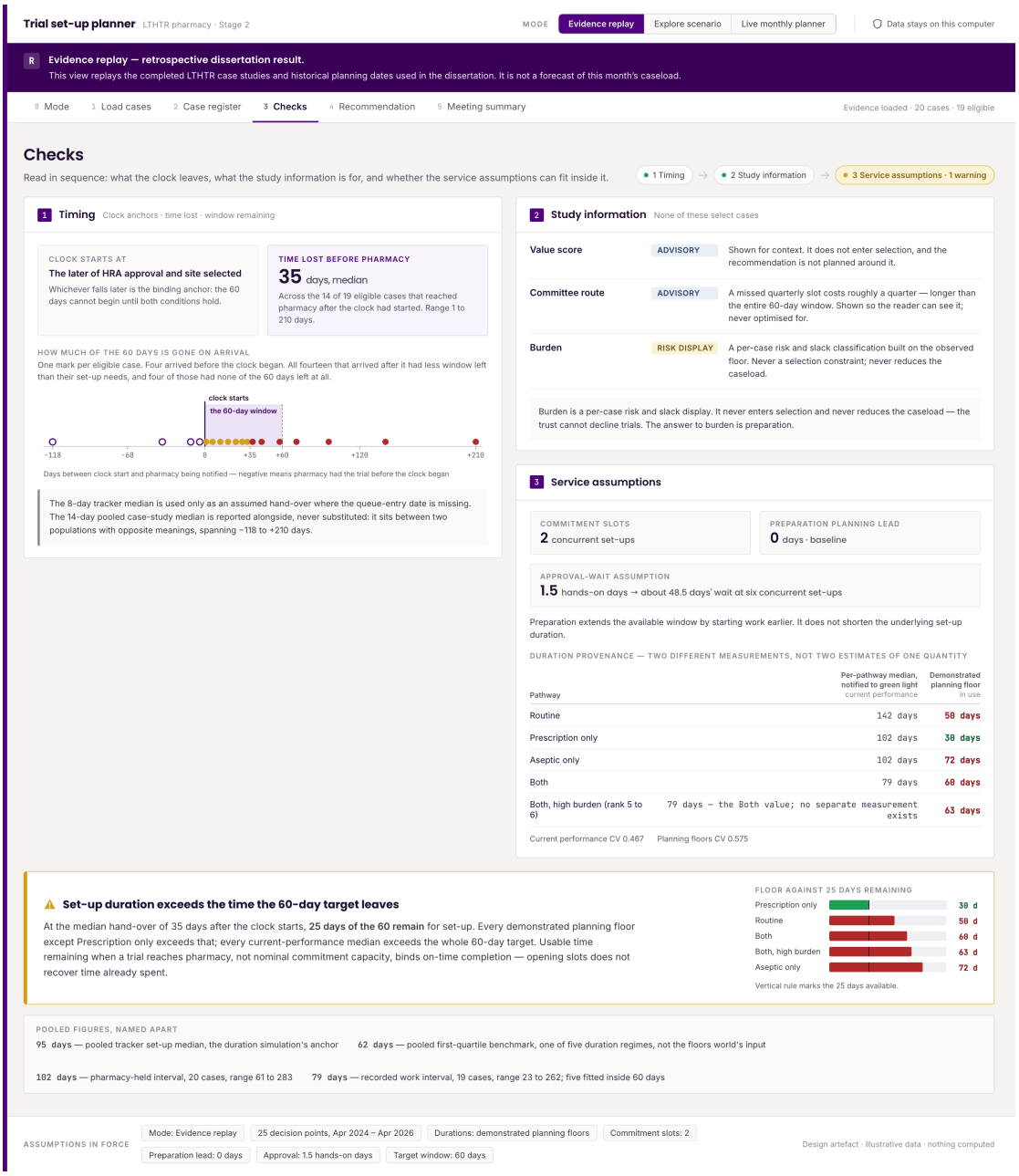


FIGURE I.4: The three checks, and the warning the third one carries. Timing gives the clock anchors and the arrival distribution, study information labels each field as advisory or as a risk display, and service assumptions set the durations in use against the twenty-five days the target leaves after a median hand-over.

I.5 The recommendation

The recommendation screen opens on the decision table in Figure I.5. Each row carries commit, defer or needs data, with the arithmetic behind it written out in a sentence a reader can check: the window is sixty days plus any early arrival or minus any time lost, set against the pathway floor. Needs data is a first-class outcome rather than an error, because a case with a missing anchor is held until the field arrives.

One point on this screen needs stating carefully, and the screen states it in a line above the table. The table covers the whole replay rather than a single month. Four trials commit across the twenty-five decision points, in four separate months; the baseline replay reaches at most one on-time trial in any one month, and three in a month is reached only at a sixty-day preparation lead, at two decision points. A reader who took the four commits as one month's plan would be reading the screen as a stronger claim than Chapter 5 makes.

Four further tabs sit behind it.

The planner records what the meeting decided. Figure I.6 shows the team's chosen plan rather than the model's. Each bar starts at that trial's own decision point, so the strip runs across the replay rather than within one month. One trial has been moved back a month to free the aseptic bench, the reason is recorded beside the row, and the consequence is stated in plain words: three of the four feasible trials finish on time instead of all four. An override is styled as a decision the tool records, not as a mistake it warns about, because the committee decides and the model advises.

The frontier shows what an extra on-time case costs. Figure I.7 plots the largest queue, 1 May 2025, at a twenty-one day preparation lead. The first on-time completion costs twelve weighted late days and the second a further one hundred and sixty-one. The recommended plan is always plotted and labelled, never the bare curve, and the on-time count is carried by height and shade together so it cannot drop out of the picture.

Slack and feasibility shows what capacity buys. In Figure I.8 the left panel gives each case its slack against the window, with the burden band as a colour overlay rather than as a ranking. The right panel is the more important half: as set-ups open from one to six the median lateness falls from sixty-one days a trial to thirty-three, and the best on-time count stays flat at one. That gap between the two curves is the finding drawn as a picture.

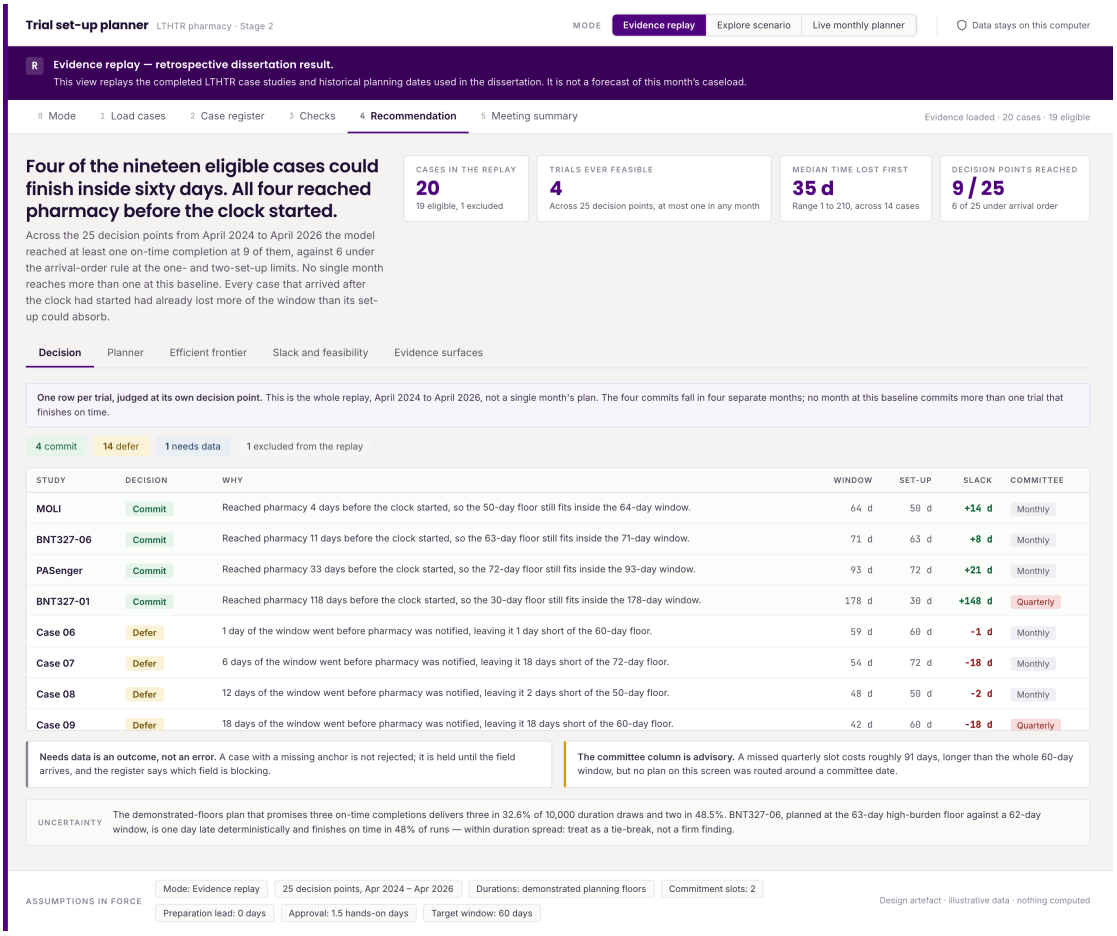


FIGURE I.5: The decision table, with the reason written out per case. Across the twenty-five decision points four trials commit in four separate months, fourteen defer, one needs data and one is excluded; the window, set-up floor and slack are given for each, and the committee badge is marked advisory.

The evidence tab holds the results the other tabs rest on. Figure I.9 pairs two rotatable surfaces with five compact panels: the historical replay at nine of twenty-five decision points against six under arrival order, the lever results, targeted against spread preparation at sixteen against twenty, the delivery simulation at 32.6 and 48.5 per cent, and the comparator sweep. The sweep is labelled research evidence only, because a rule that wins in one duration regime of five is not advice to give a service planning next month.

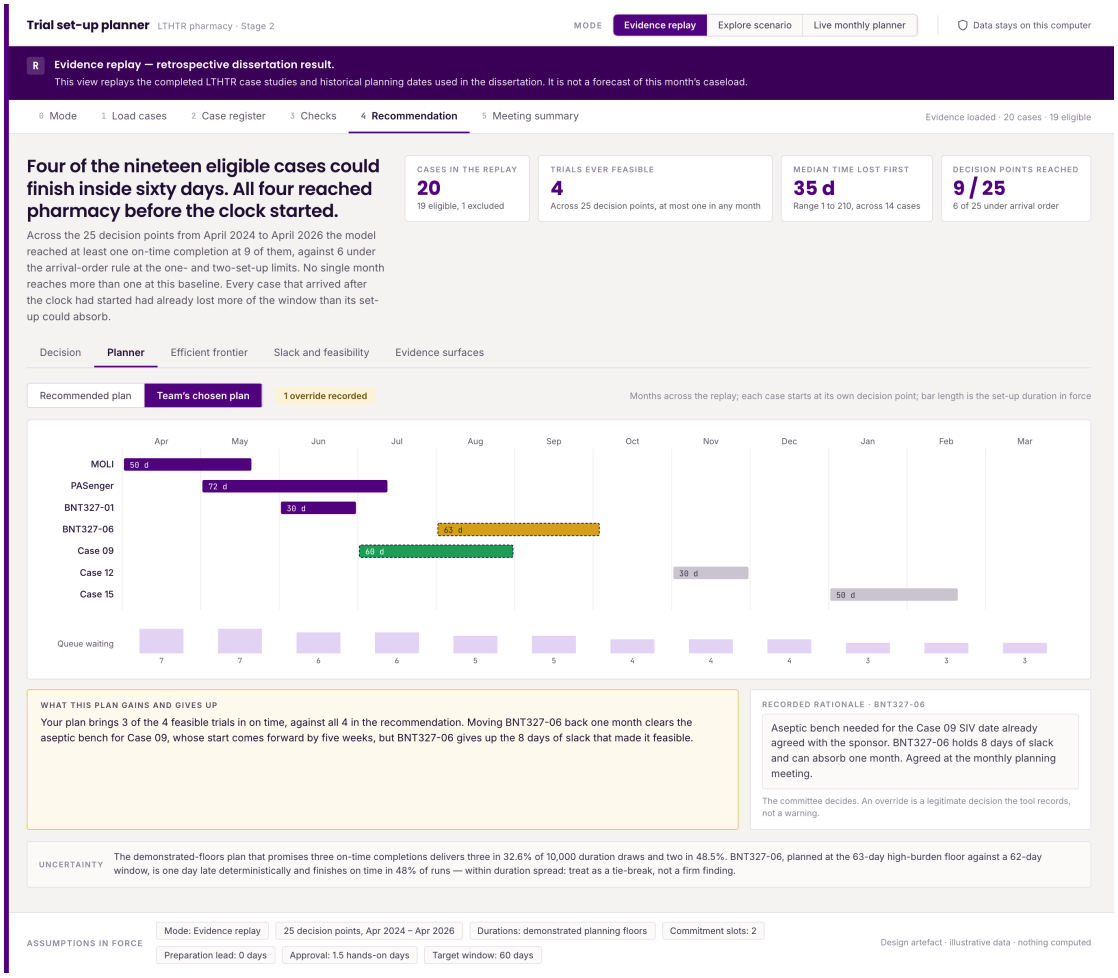


FIGURE I.6: The planner, showing an override and what it costs. Bars run from each projected start with the queue burn-down beneath; the moved case and the case that gains from the move are marked, the recorded rationale sits beside them, and the consequence is stated as three of the four feasible trials on time against all four in the recommendation.

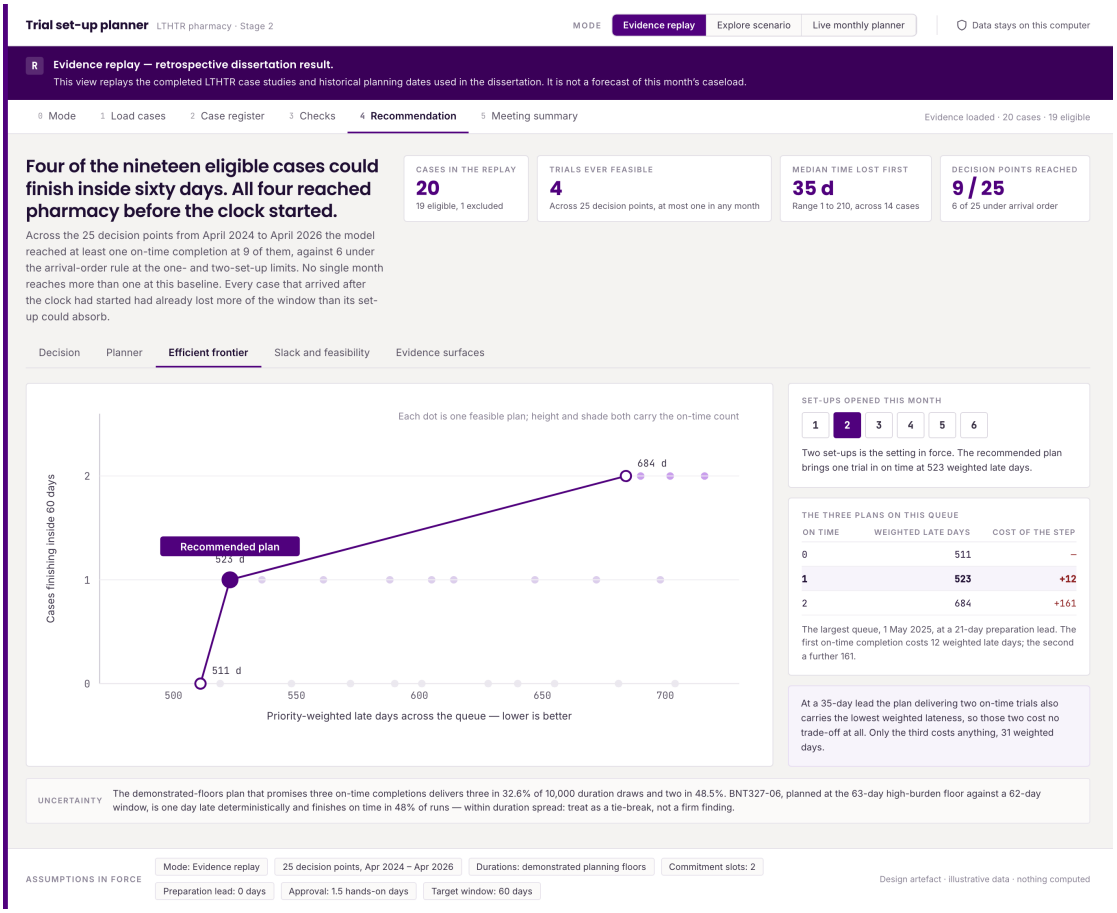


FIGURE I.7: The frontier, with the recommended plan labelled. Feasible plans are shown as a cloud with the frontier drawn along the boundary, the recommended plan marked, and the three frontier plans tabulated with the weighted late days each step costs.

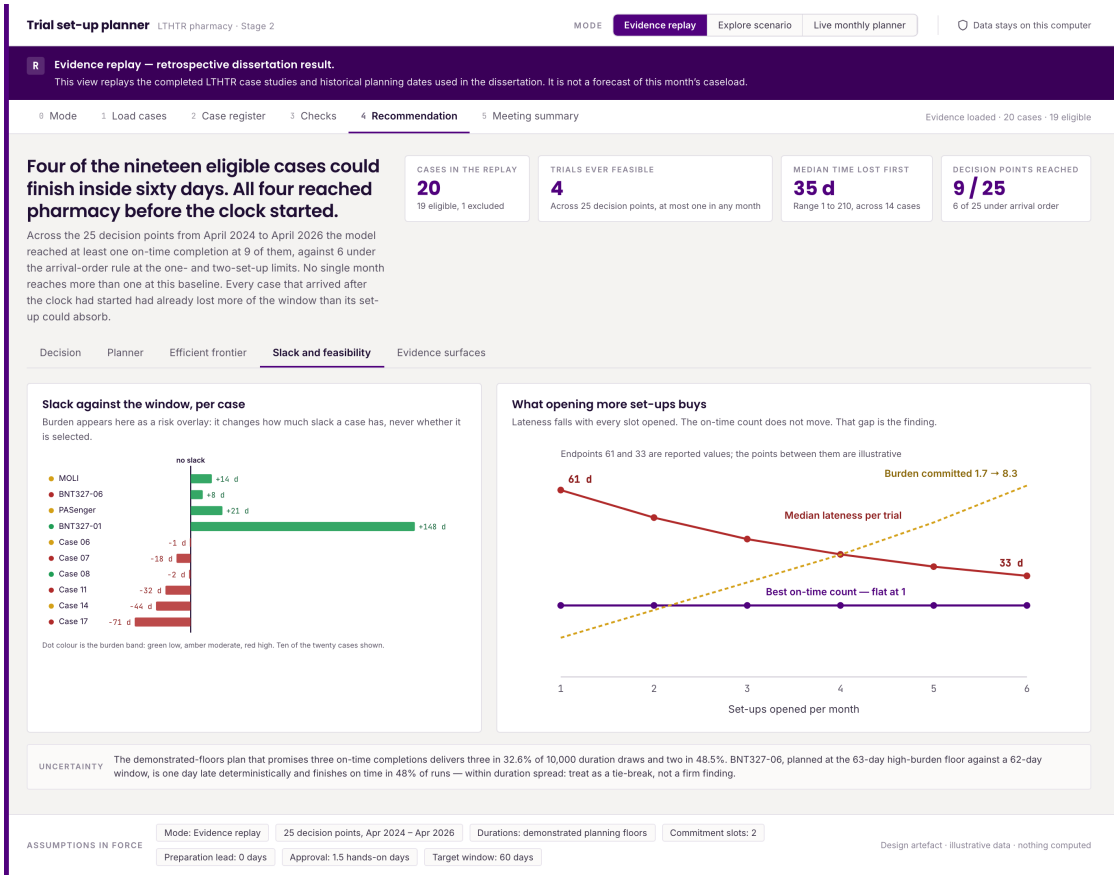


FIGURE I.8: Slack per case, and what opening more set-ups buys. Slack is plotted against the window with burden shown as a colour overlay, and lateness, on-time count and committed burden are plotted against set-ups opened per month.

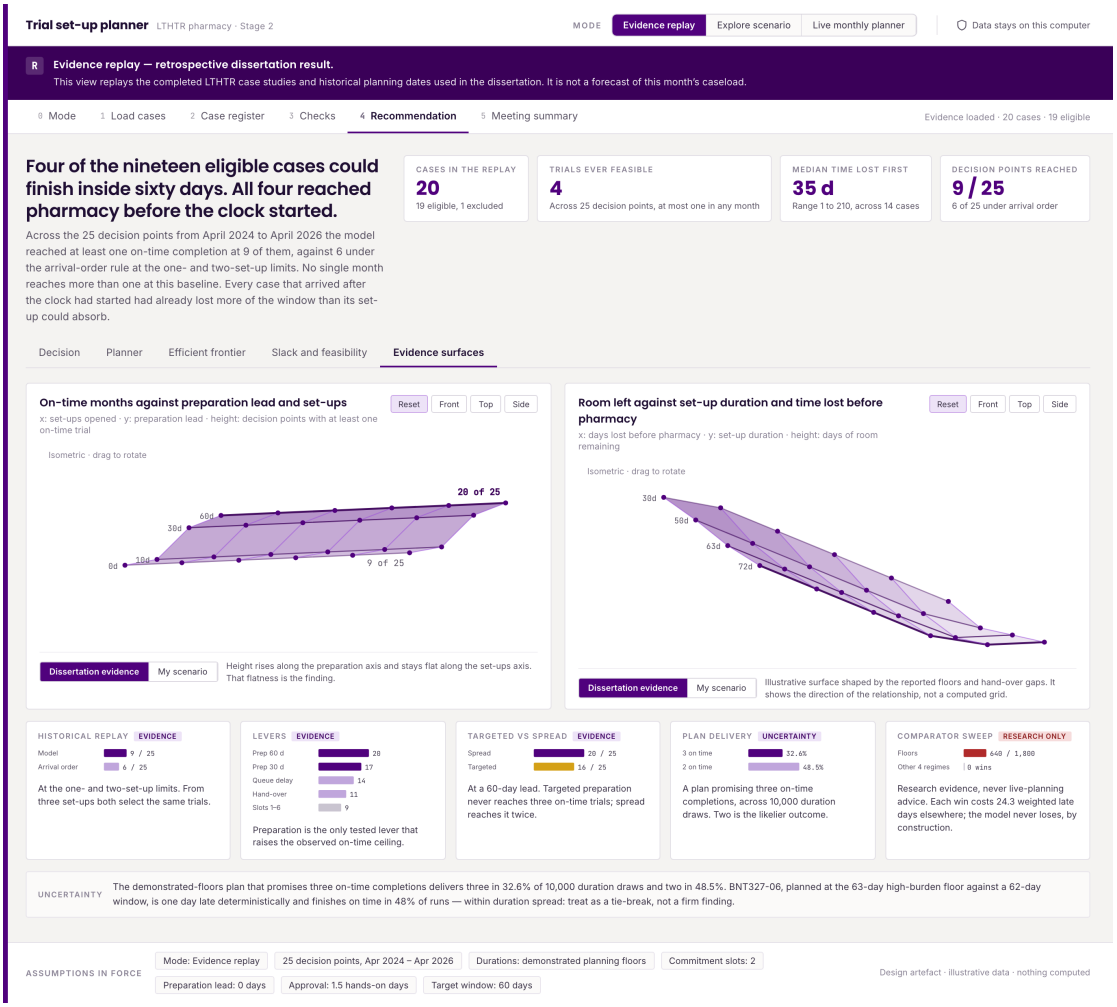


FIGURE I.9: The evidence tab, with two surfaces and five supporting panels. The first surface rises along the preparation axis and stays flat along the set-ups axis; the panels carry the replay comparison, the lever results, targeted against spread preparation, the delivery simulation and the comparator sweep.

I.6 The meeting summary

Figure I.10 is one page of A4 to take into the meeting. It carries the mode and its caveat, the decisions with their reasons, the assumptions the plan rests on, and a list of the values that were missing or assumed. The last of these matters most on paper, because a printed page loses the colour coding that marks an assumed value on screen.

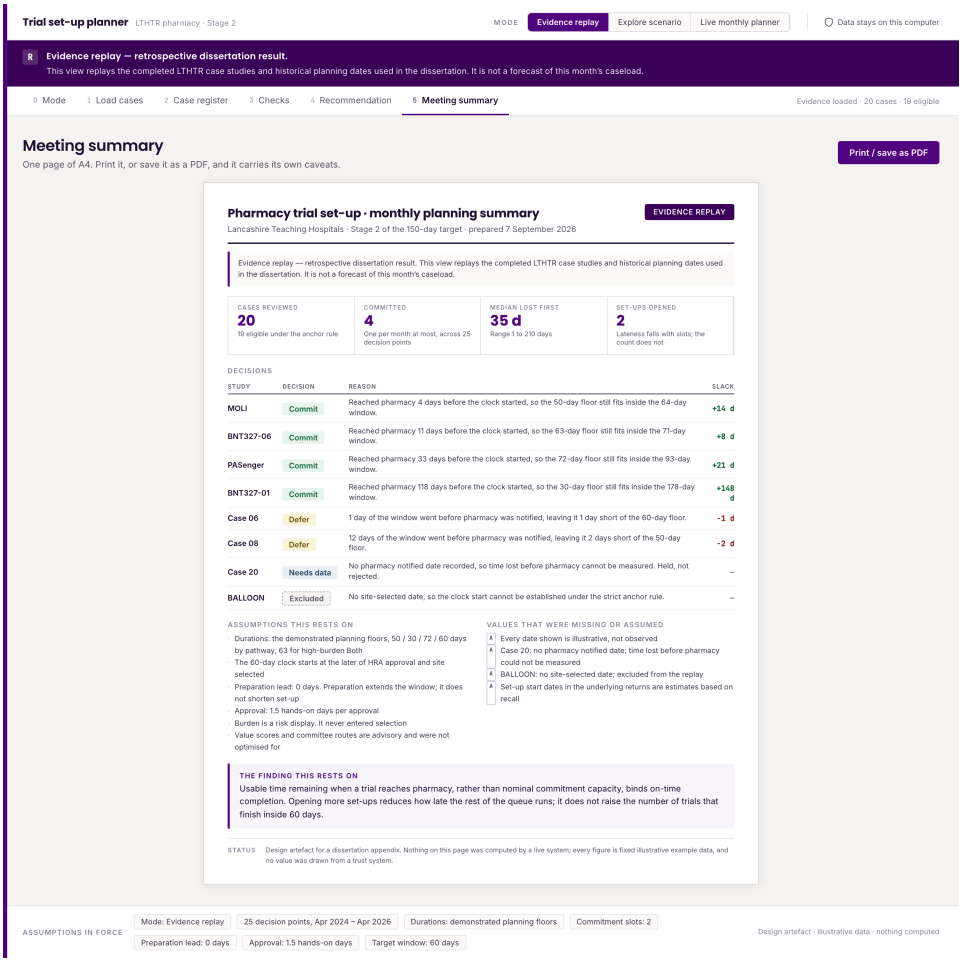


FIGURE I.10: The printable meeting summary. One A4 page carries the mode caveat, the decision list with reasons and slack, the assumptions in force, the values that were missing or assumed, and a statement of the finding.

I.7 An implementation pathway

Table I.1 sets the recommended actions of Section 6.4 against the owner, the minimum data each needs, a measure of success and its precondition, with a prospective pilot to calibrate the model against realised outcomes.

Table I.1: Implementation pathway for the recommended actions

Action	**Owner**	**Minimum data**	**Success measure**	**Condition**
Record approval time	Pharmacy CT lead	Approval-ready and green-light timestamps, trial ID	Observed approval-time distribution after 3–6 months	Must separate waiting from hands-on approval
Flag aseptic studies early	R&I + Pharmacy	Aseptic requirement, notification date	Share notified before the clock starts	Risk screening, not a value priority
Monthly feasibility screen	Pharmacy + R&I	Queue entry, pathway, clock anchor, lead	On-time feasibility and queue lateness	Supports judgement, does not replace governance
Commit to a reliable outcome	Pharmacy + R&I	Duration distribution, selected plan	Share of plans delivering the stated count	Requires a stated risk appetite
Prospective pilot	Service management	Decision logs, completion dates	Calibration error, on-time rate	Not causal without a comparator design

I.8 What the mock-up does not do

Four limits are worth stating plainly.

It computes nothing. Every figure on the screens is fixed, so the interface demonstrates a layout and a set of safeguards rather than a working decision aid.

It has not been used. No member of the Research and Innovation team or Pharmacy Clinical Trials has planned a month with it, so nothing here is evidence that the screens are usable, only that they are specified.

Its case rows are illustrative. Five real study names appear, which the dissertation already publishes; the rest are numbered cases, and every date is illustrative and marked as such. Only the aggregate figures on the recommendation and evidence screens are reported results.

Its outputs would be conditional. Even in live use, the recommendation is a plan under the assumptions listed at the foot of the screen, and Section 5.3 is the reason the interface says so on every tab: a plan promising three on-time completions delivers three in about a third of simulated runs, and a margin smaller than the duration spread is a tie-break rather than a finding.

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